



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

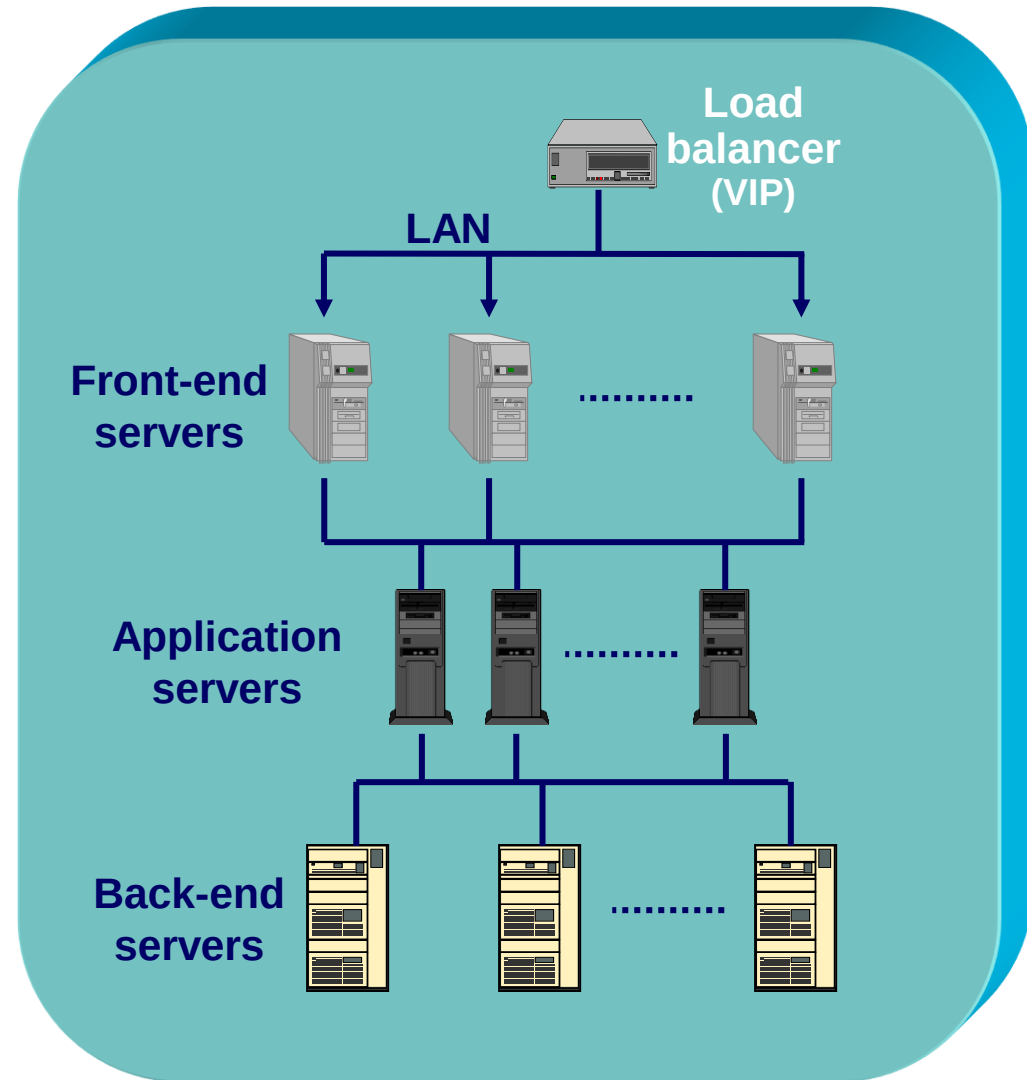
Geographical replication

Michele Colajanni

Dipartimento di Informatica – Scienza e Ingegneria
michele.colajanni@unibo.it

Third question

- 1) How many servers can we add?
- 2) How many servers it is convenient to add?
- 3) *Where to place it?*



Placement



Where to place you IDCs?



Factors influencing where to place data centers

Placement and sizing of an IDC is an interesting and challenging optimization problem including **technical** and **management** factors:

1. Zoning

- cheap land and space for future expansions
- safe land
- proximity to stores operating 24 hours a day

2. Networking

- presence of large POPs
- excellent (multiple) backbones

3. Energy

- proximity to abundant and cheap electrical services
- power concessions

4. County and city policies

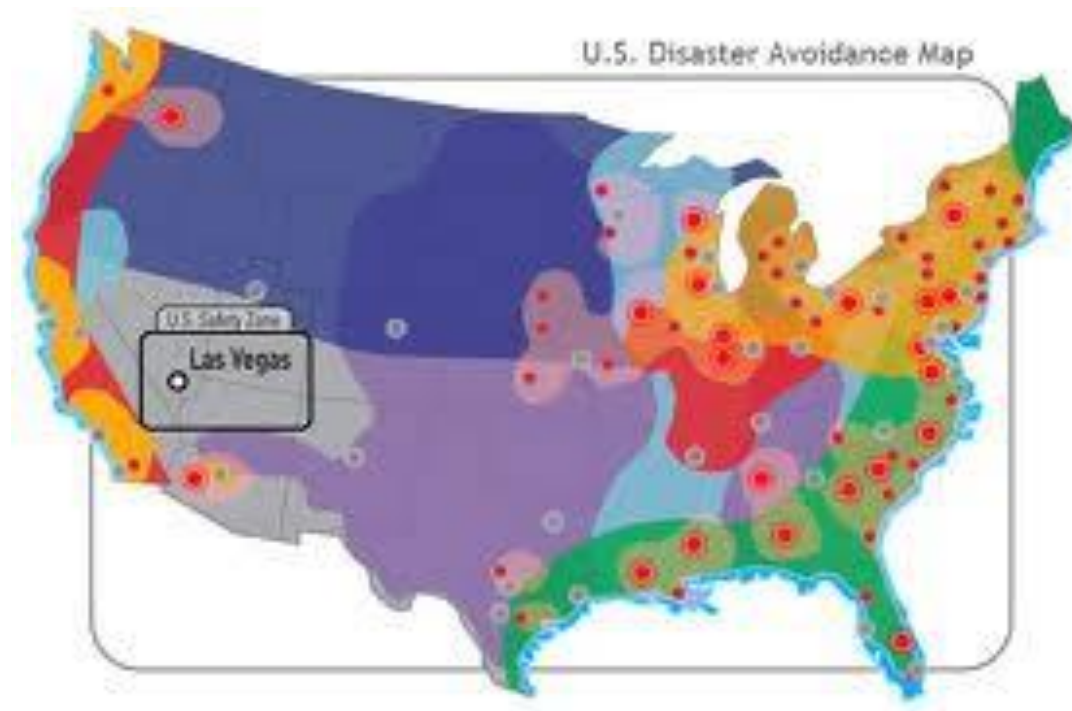
- incentives
- tax break policies

5. Availability of human competences



Zoning: safe land

The datacenters should be located in the #1 rated geographically safety zones



Networking: IDCs tend to flock together

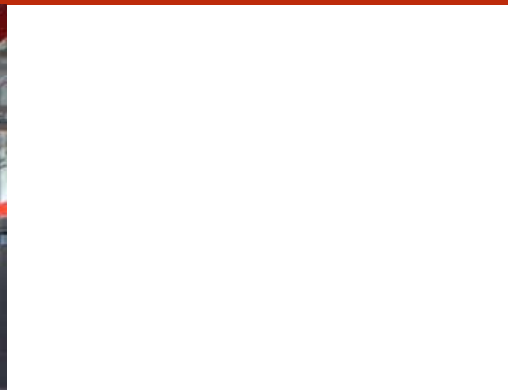
- The interesting but unsurprising fact about datacenters is that **they tend to be located in the same areas**
- There will usually be some anonymous buildings containing millions of dollars of hardware
- So when the big players of the world will decide to hook up, they will find that it is relatively cheap: “often their counterparts are just across town”



Top 20 hosting cities



Ex., SuperNAP near Las Vegas

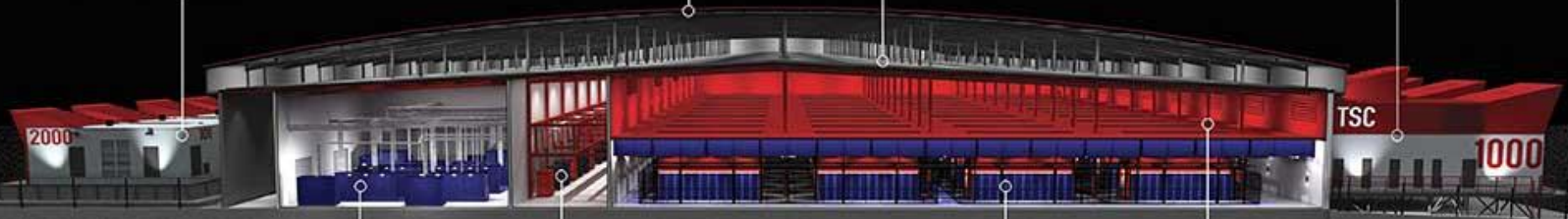


SWITCH SHIELD 1
INDEPENDENT 200 MPH
STEEL DATA CENTER ROOF
100% PENETRATION-FREE

SWITCH SHIELD 2
INDEPENDENT 200 MPH
STEEL DATA CENTER ROOF
100% PENETRATION-FREE

WDMD SWITCHMOD 2000
WATTAGE DENSITY MODULAR DESIGN
CONTINUOUS RATED GENERATOR PLANTS

TSC 1000 ROTOFLY
UNINTERRUPTABLE, MULTI-SYSTEM
HVAC DATA CENTER COOLING UNITS



EACH 10 MVA POWER SYSTEM
CONSISTS OF 11 SEPARATED AND
COMPARTMENTAUZED FIRE-RATED
CONCRETE AND STEEL POWER ROOMS

THE SUPERNAP POWER SPINE IS A QUAD
DIVIDED PDU CORRIDOR THAT ENABES
UNEQUALED DENSITY WHILE MAINTAINING
MAINTENANCE AND SECURITY SEPARATIONS

THE SUPERNAP PROPRIETARY T-SCIF HEAT
CONTAINMENT STRUCTURES ALLOW FOR ANY
TYPE OF GEAR WHILE PROVIDING THE INDUSTRY'S
HIGHEST LEVEL OF ENERGY EFFICIENCY

THE SUPERNAP HEAT CEILING ENABES 100%
HEAT CONTAINMENT ABOVE THE THERMAL
SCIF PODS AS WELL AS AN INSULATED
COMPARTMENTALIZED PATH TO THE TSC UNITS

Energy sources: *examples*



Google in Oregon



Google in Dallas



Why incentives and competition among cities?

AWS Has Spent \$35 Billion on its Northern Virginia Data Centers

Amazon Web Services (AWS) has spent \$35 billion on its cloud computing infrastructure in Northern Virginia over the past 10 years, illustrating the enormous impact data centers can have on regional economies.

[Rich Miller](#)

Oct. 4, 2021

Google to spend \$9.5B this year on data centers and offices

Investing in offices is "more important than ever," Google CEO Sundar Pichai said, even as the company embraces hybrid work



Site - AVOID

- Risk of seepage water from outside or from broken pipes indoor
→ Ancient and wrong choice: to locate the datacenter at ground level or even below)
- Vapor and moisture infiltration
- Direct irradiation of sunlight
- Not monitored entrances through other areas
- Adjacency to tanks and diesel generators
- Lack of facilities for logistics, truck access, lifts, elevators, etc.



Typical IDC issues in one year

- ~0.5 overheating (power down most machines in <5 mins, ~1-2 days to recover)
- ~1 PDU failure (~500-1000 machines suddenly disappear, ~6 hrs to come back)
- ~1 rack-move (plenty of warning, ~500-1000 machines powered down, ~6 hrs)
- ~1 network rewiring (rolling ~5% of machines down over 2-day span)
- ~20 rack failures (40-80 machines instantly disappear, 1-6 hours to get back)
- ~5 racks go wonky (40-80 machines see 50% packet loss)
- ~8 network maintenances (4 might cause ~30-minute random connectivity losses)
- ~12 router reloads (takes out DNS and external VIPs for a couple minutes)
- ~3 router failures (have to immediately pull traffic for an hour)
- ~dozens of minor 30-second blips for DNS
- ~1000 individual machine failures (2-4% failure rate, machines crash at least twice)
- ~thousands of hard drive failures (1-5% of all disks will die)

In addition, software bugs, configuration errors, human errors, ...



Tips for performance and resiliency

1. Check for bottleneck
2. Check for single points of failure
3. Keep it simple and then scale (**KISS**): the reason many systems use centralized control
4. If it is not well tested, do not rely on it
 - *Question:* how do you test availability with hundreds of loosely coupled services running on thousands of machines?



In summary

Technique	Performance	Availability
Replication	✓	✓
Partitioning (sharding)	✓	✓
Load-balancing	✓	
Watchdog timers		✓
Integrity checks		✓
Canaries		✓
Eventual consistency	✓	✓



1st Trend → Mega data-centers

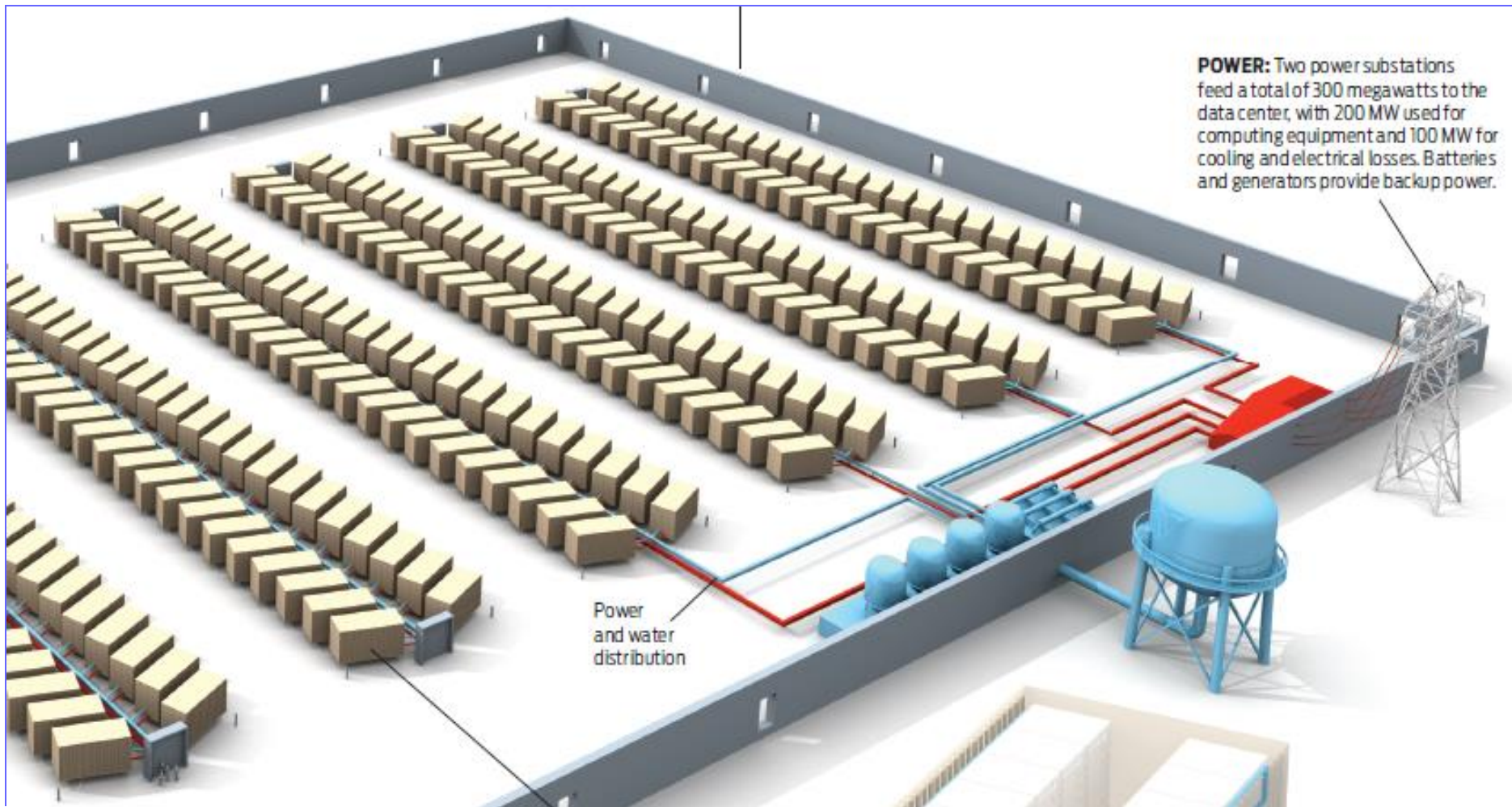
- Large providers need some number of mega-datacenters to house large computations
- The size of a mega-datacenter is typically determined by **extracting the maximum benefit from the economies of scale** available at the time the data center is designed
- This is an exercise in jointly optimizing server cost and power availability:

Typical design →

100,000s of servers spread over 100,000s of square feet drawing from 10 to 20MW of power



“The million server datacenter”



Geographical replication

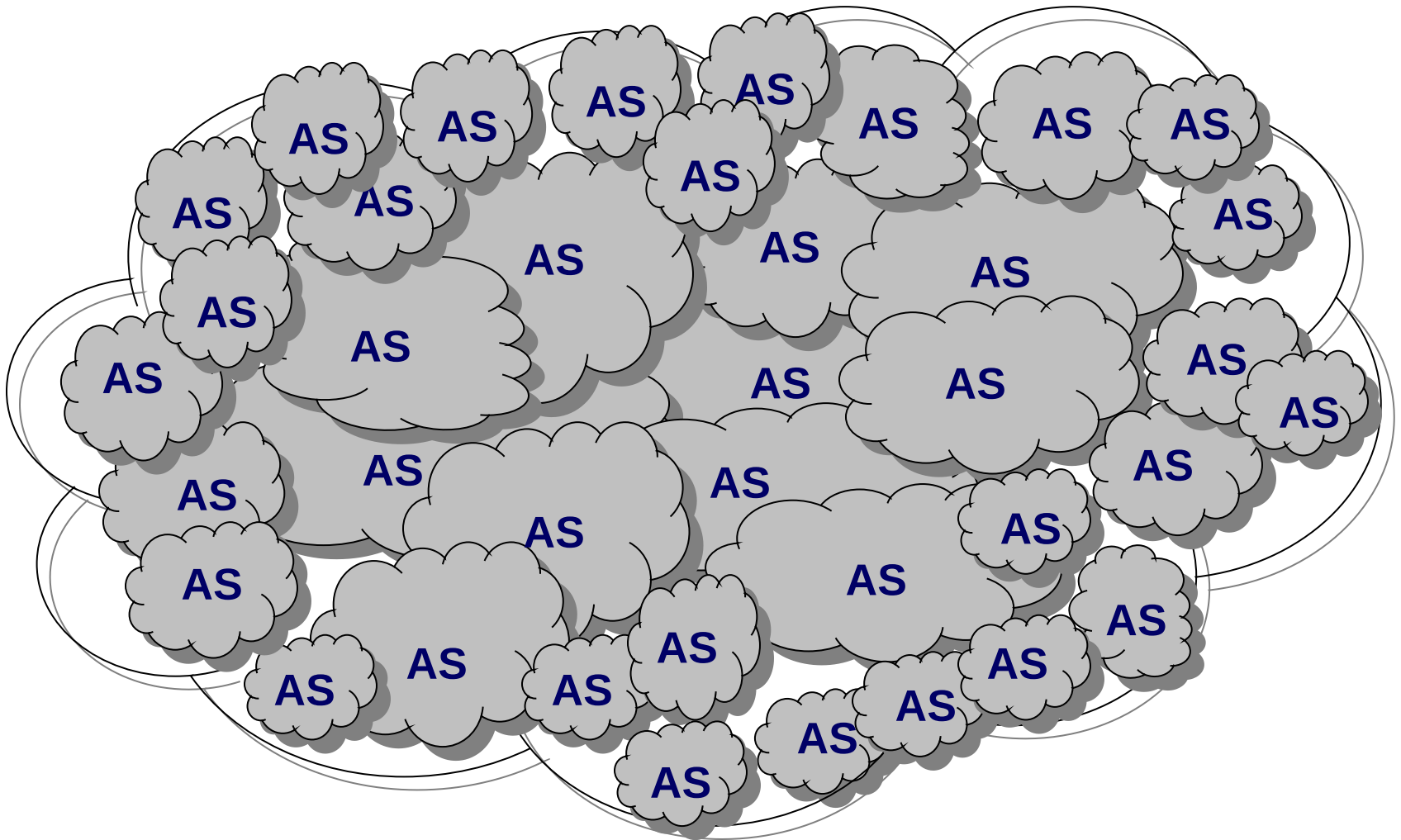


Questions related to geo-replication

- There are also challenges to address at the design level:
 - **Where to place servers?**
 - **Where to place primary node?**
 - **Where to place secondary nodes?**
 - **How big to make each node?**
1. Few **mega-datacenters**?
 2. Many regional datacenters (also called **mini or micro-datacenters**)?
 3. Combinations of the two?



What's Internet?

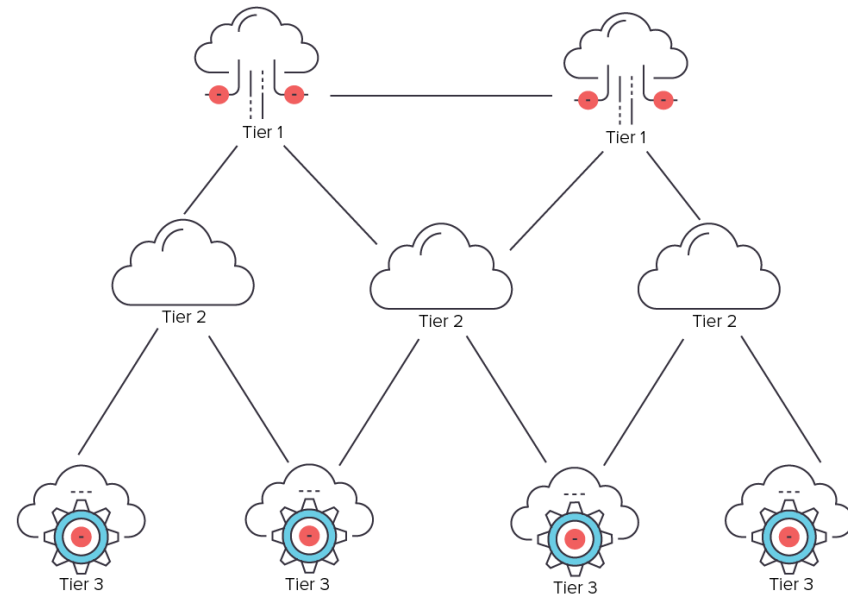


A system of interconnected Autonomous Systems



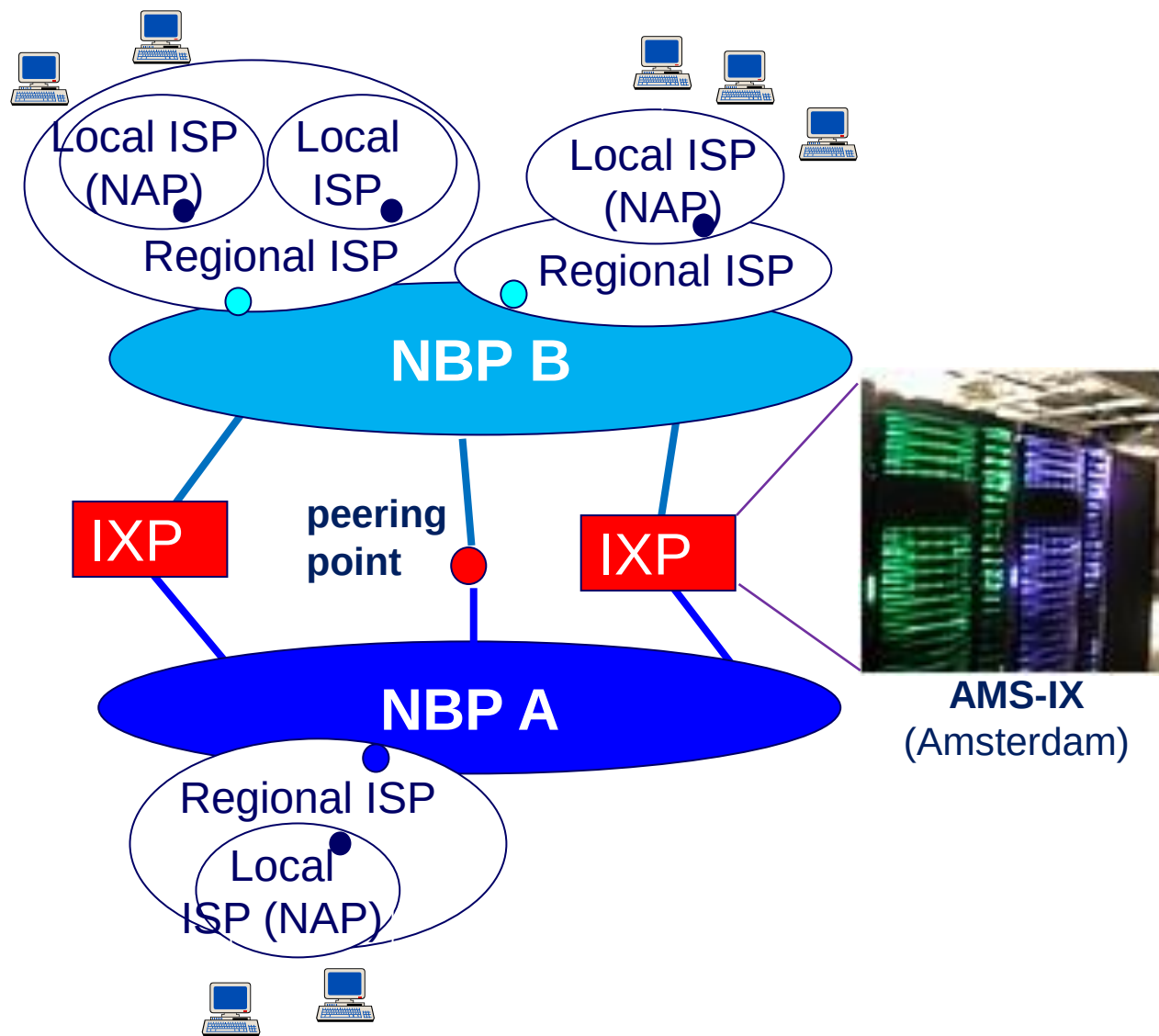
Tier 1 ISP

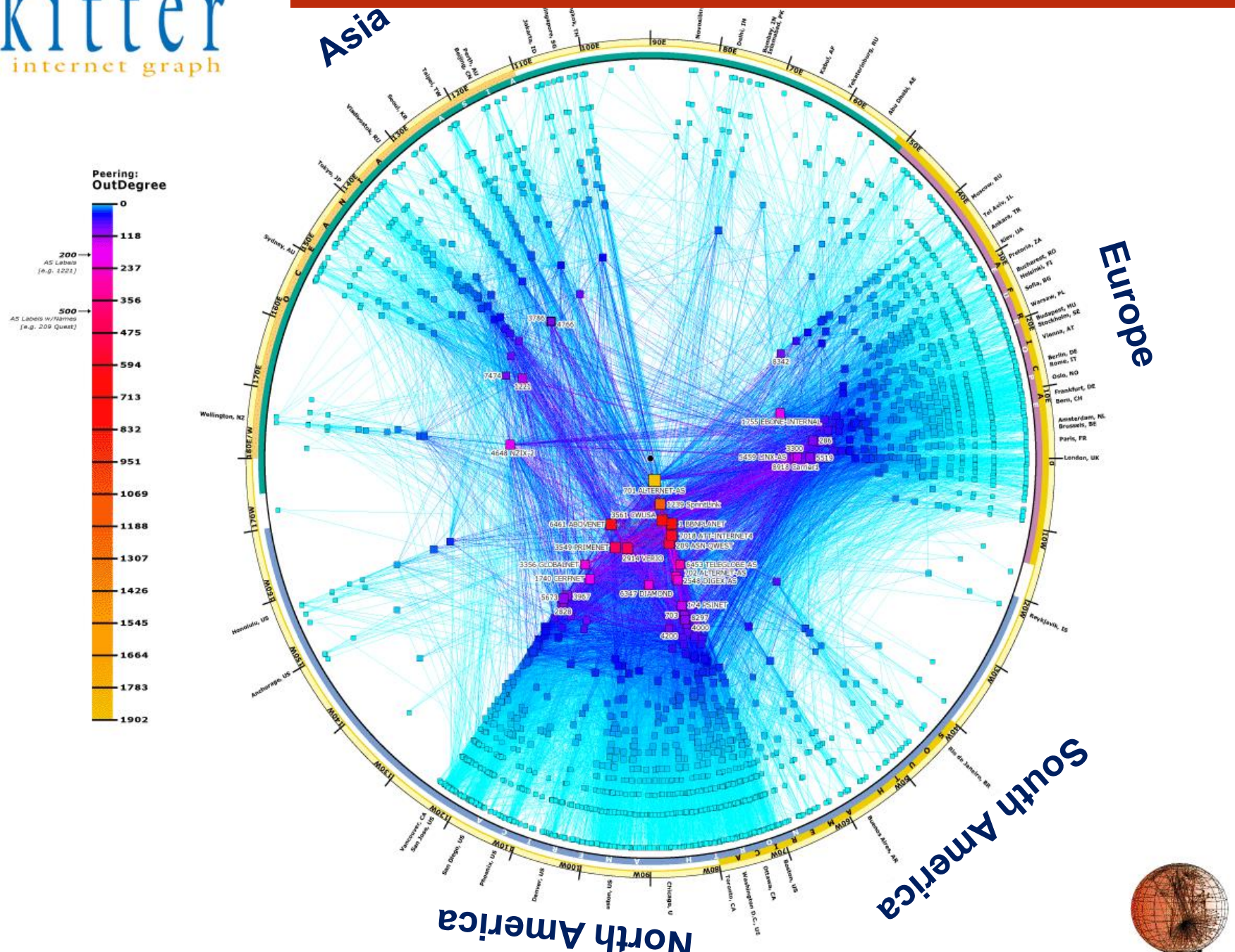
- 1) AT&T Services (AS7018), US
- 2) Century Link (AS3356), US
- 3) Deutsche Telekom (AS3320), Germany
- 4) GTT Communications (AS3257), US
- 5) Liberty Global (AS6830), US-UK-NL
- 6) NTT Communications (AS2914), Japan
- 7) Orange (AS5511), France
- 8) PCCW Global (A3491), Hong Kong
- 9) Sprint Nextel Corporation (AS1239), US
- 10) TATA Communications (AS6453), India
- 11) Telecom Italia Sparkle (AS6762), Italy
- 12) Telxius (AS12956), Spain
- 13) Telia Carrier (AS1299), Sweden
- 14) Verizon Business (AS701), US
- 15) Zayo Group (AS6461), US



You have also to take into account how ASes are interconnected

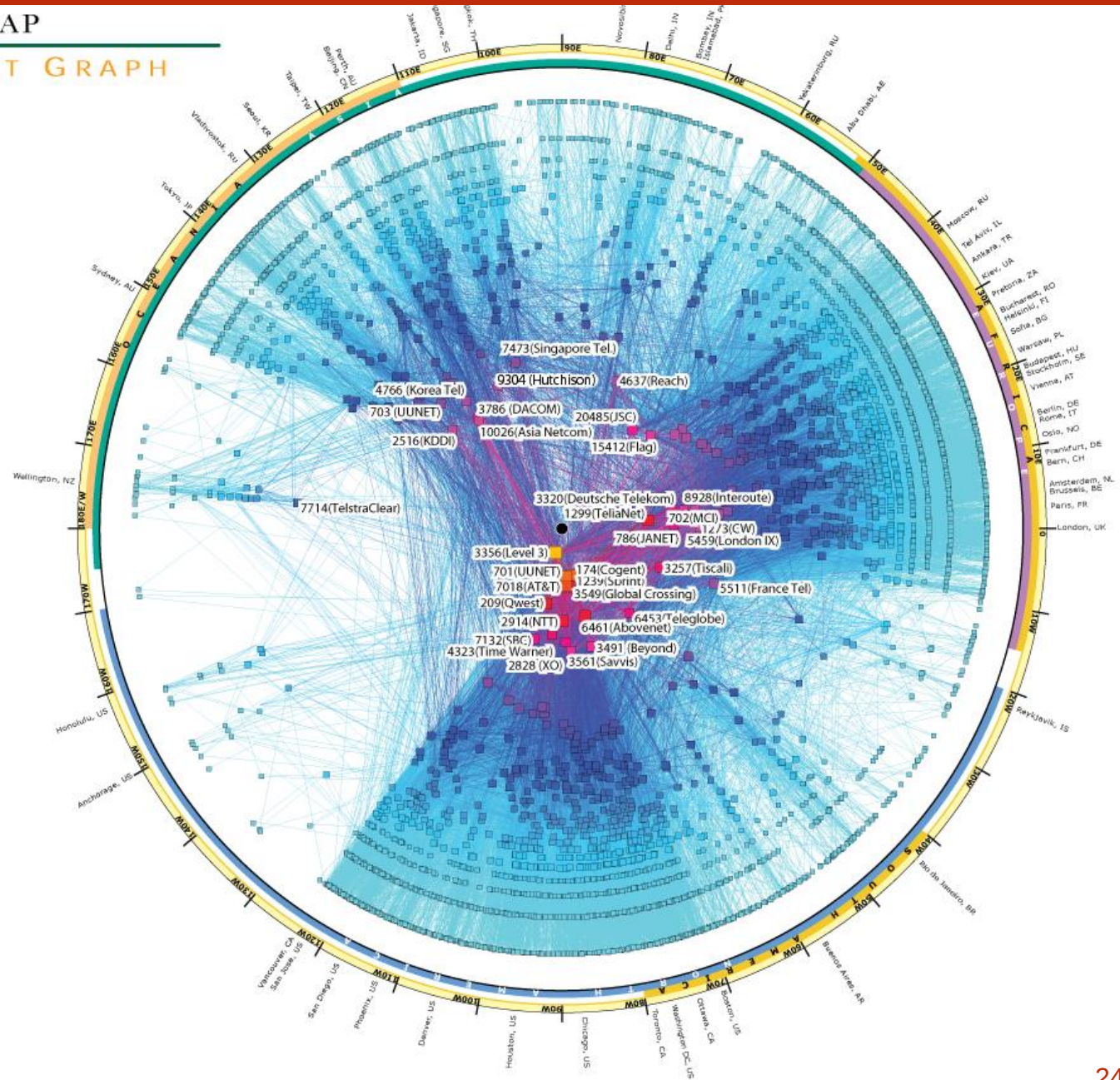
- **peering point**
(private)
- **Internet Exchange Point (IXP or IX)**



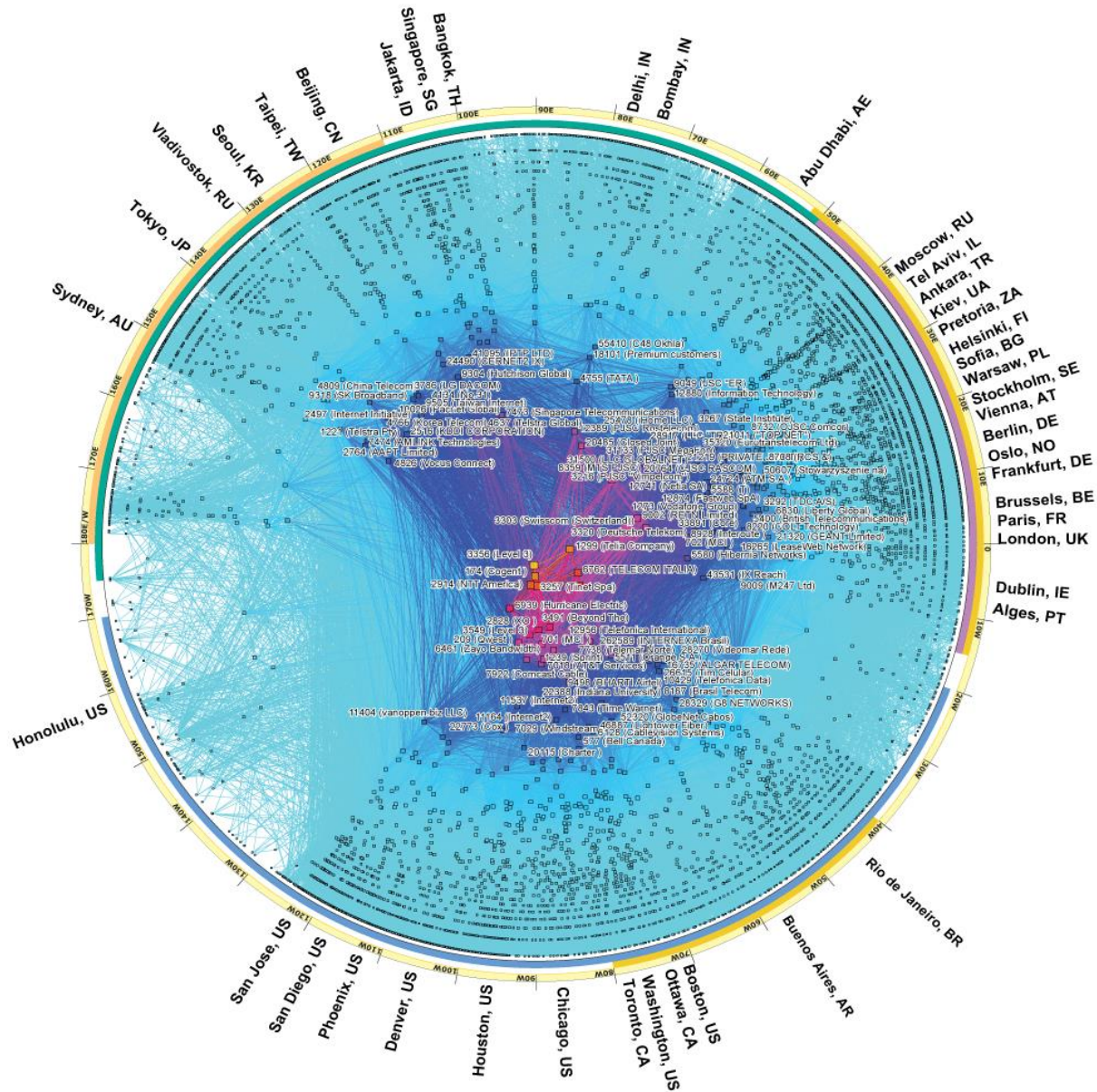


IPv4 Internet TOPOLOGY MAP

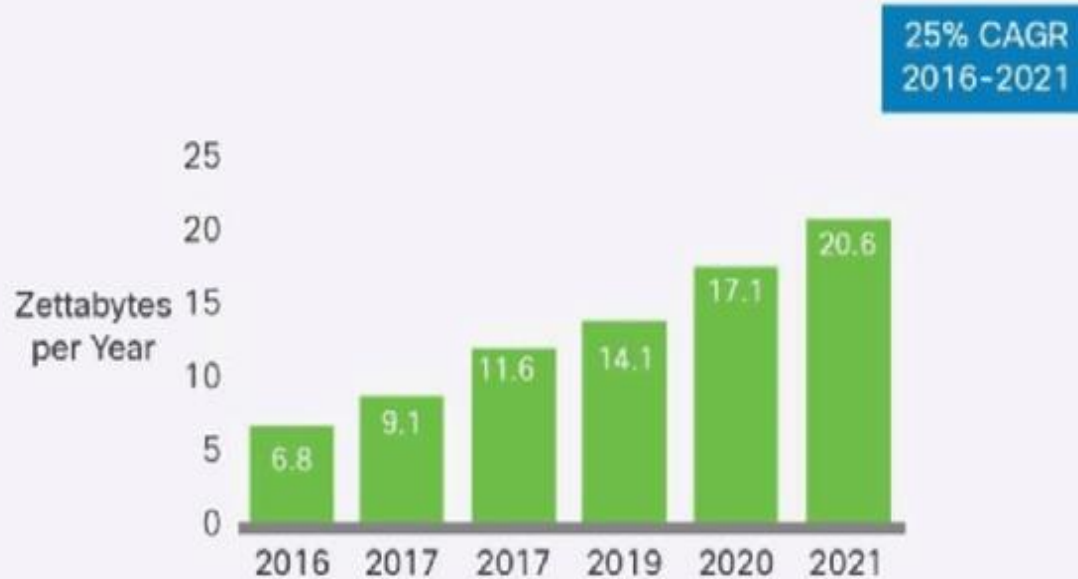
AS-level INTERNET GRAPH



Interconnections among AS (2017)



Data Center traffic (*Annual Growth Rate*)

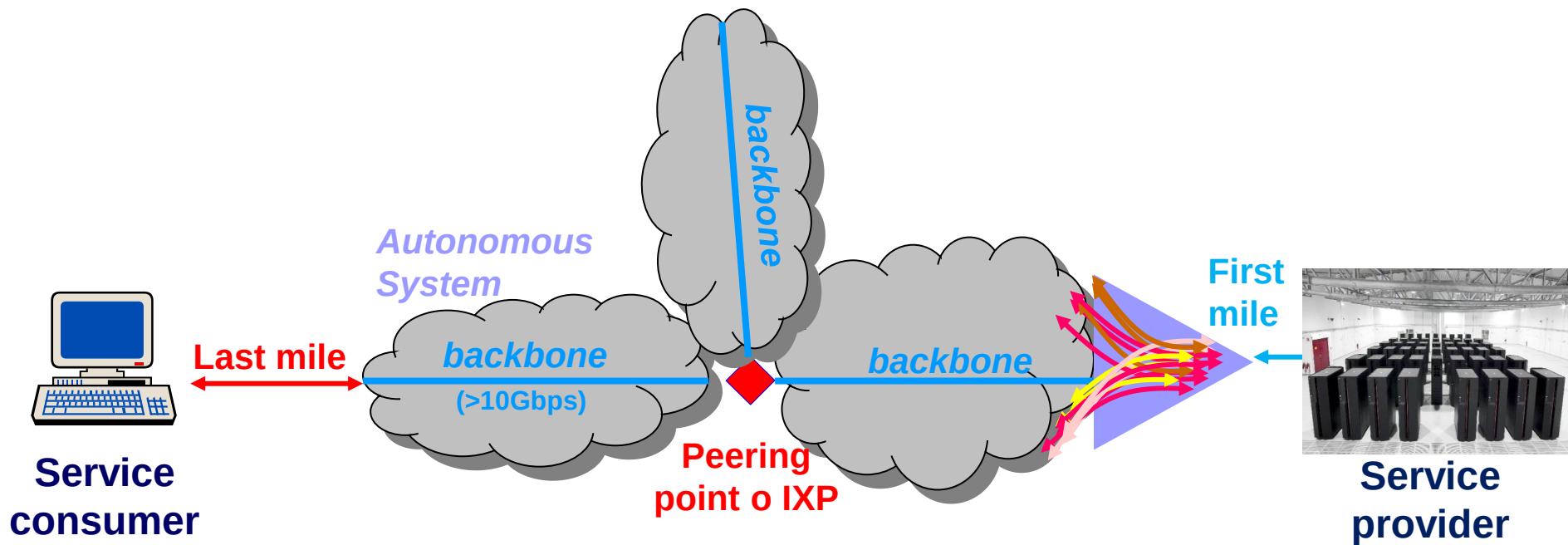


Source: Cisco Global Cloud Index, 2016-2021.

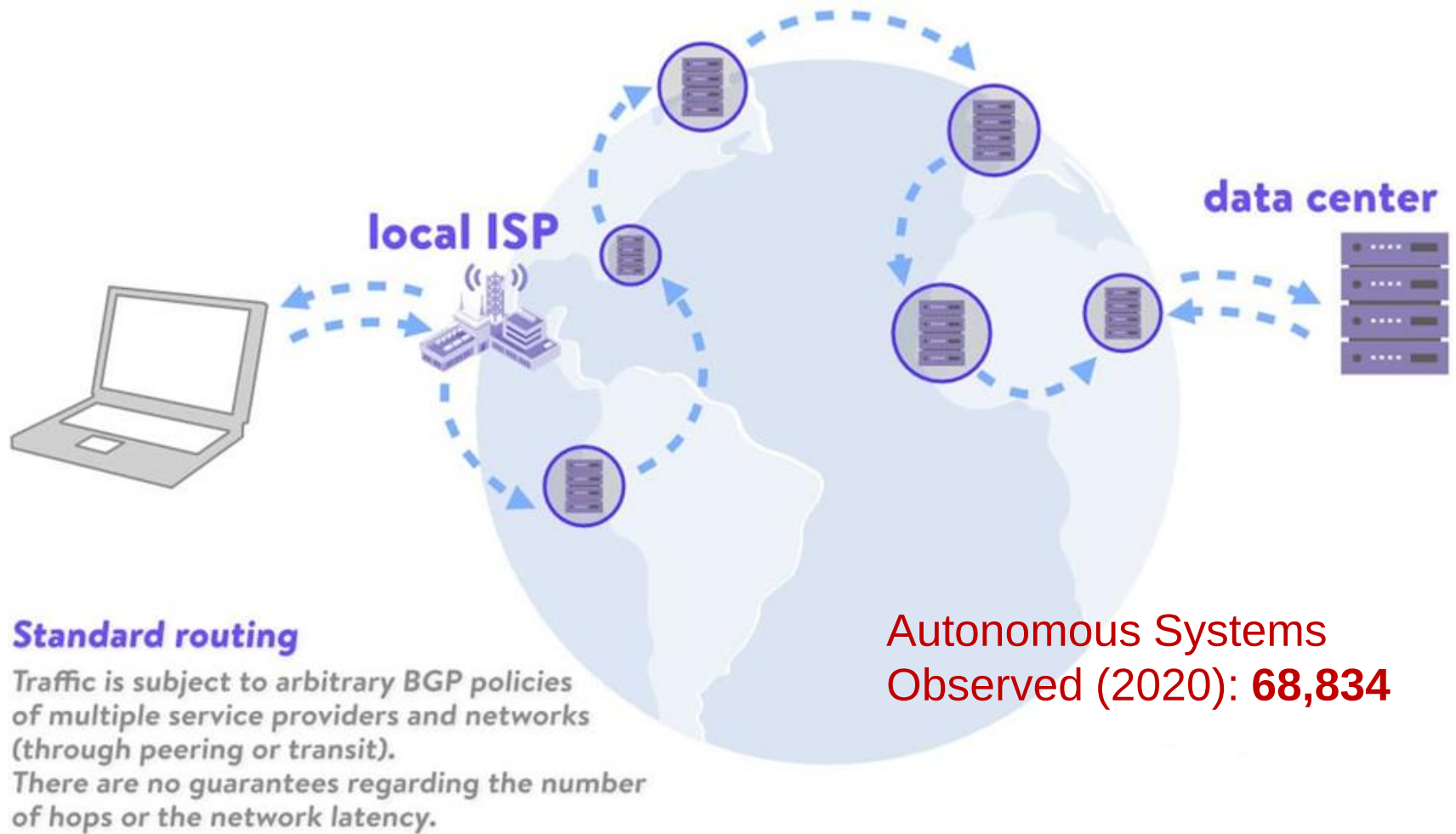
1000 **kilobytes** = 1 Megabyte
1000 **Megabytes** = 1 Gigabyte
1000 **Gigabytes** = 1 Terabyte
1000 **Terabytes** = 1 Petabyte
1000 **Petabytes** = 1 Exabyte
1000 **Exabytes** = 1 Zettabyte
1000 **Zettabytes** = 1 Yottabyte
1000 **Yottabytes** = 1 Bronobyte
1000 **Bronobytes** = 1 Geopbyte



The importance of network connection



The long path towards the “cloud”



The network impact

1. Last mile problem

2. Network traffic

3. Congestion near the provider site

- As the number of users increases, services can become congested, similar to what happens to roads during rush hour. At this point, the scalability actually decreases with the number of users, instead of increasing
- Cloud providers must take care to make sure that the congestion point is greater than the expected usage level by
 - increasing available capacity
 - replicating service points



The network impact close to the *provider*

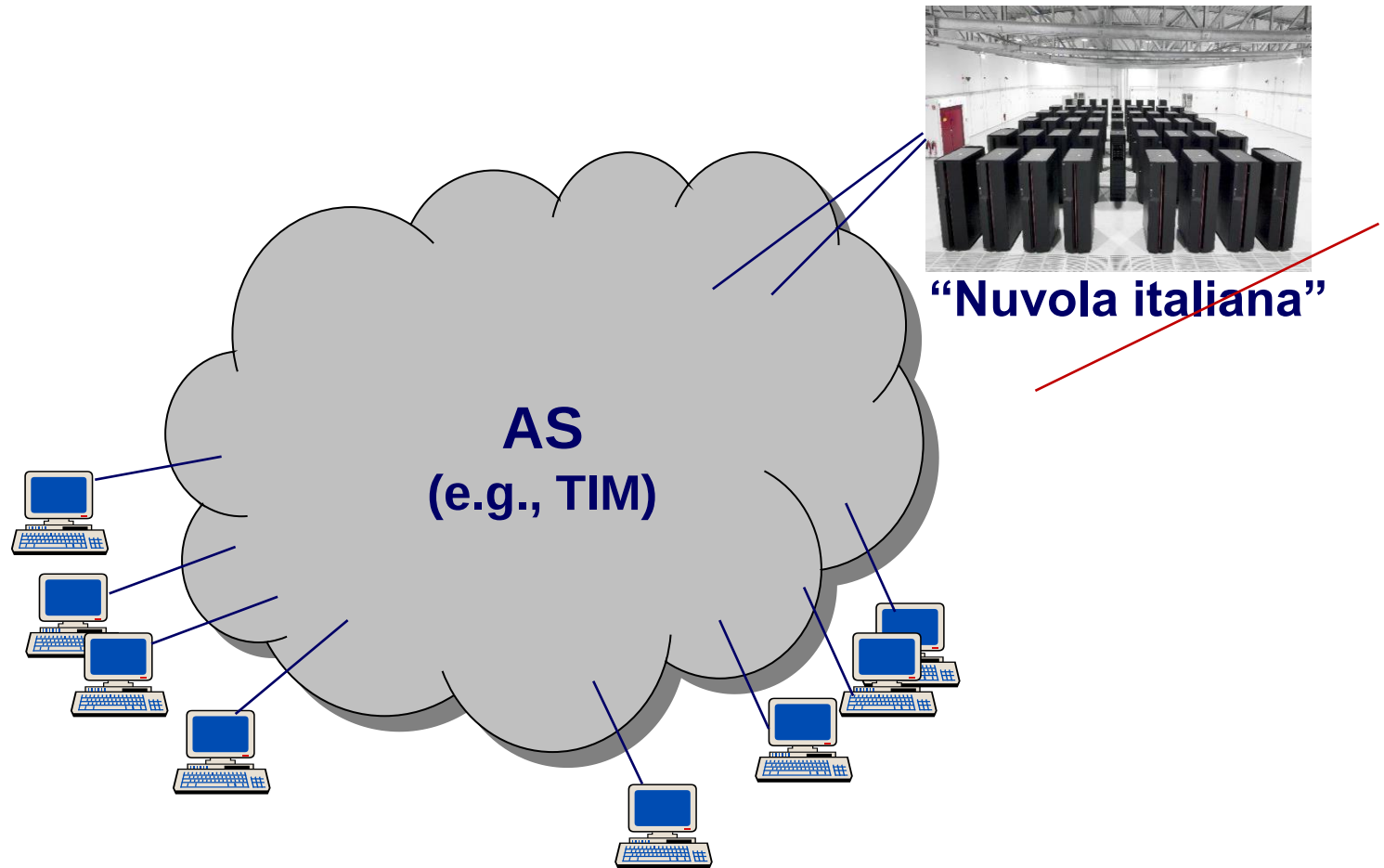
1. First mile problem

2. Network congestion near the provider site

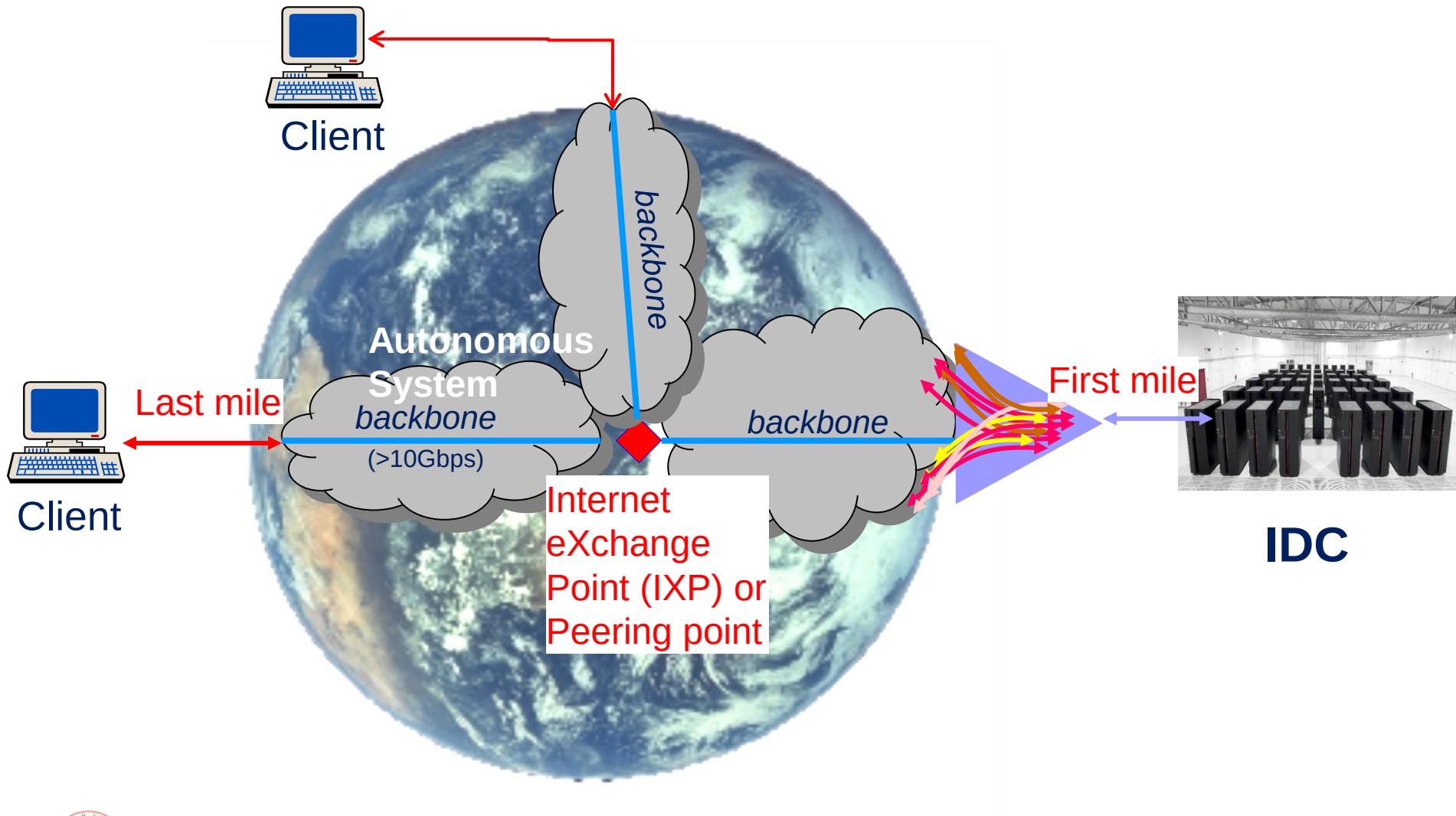
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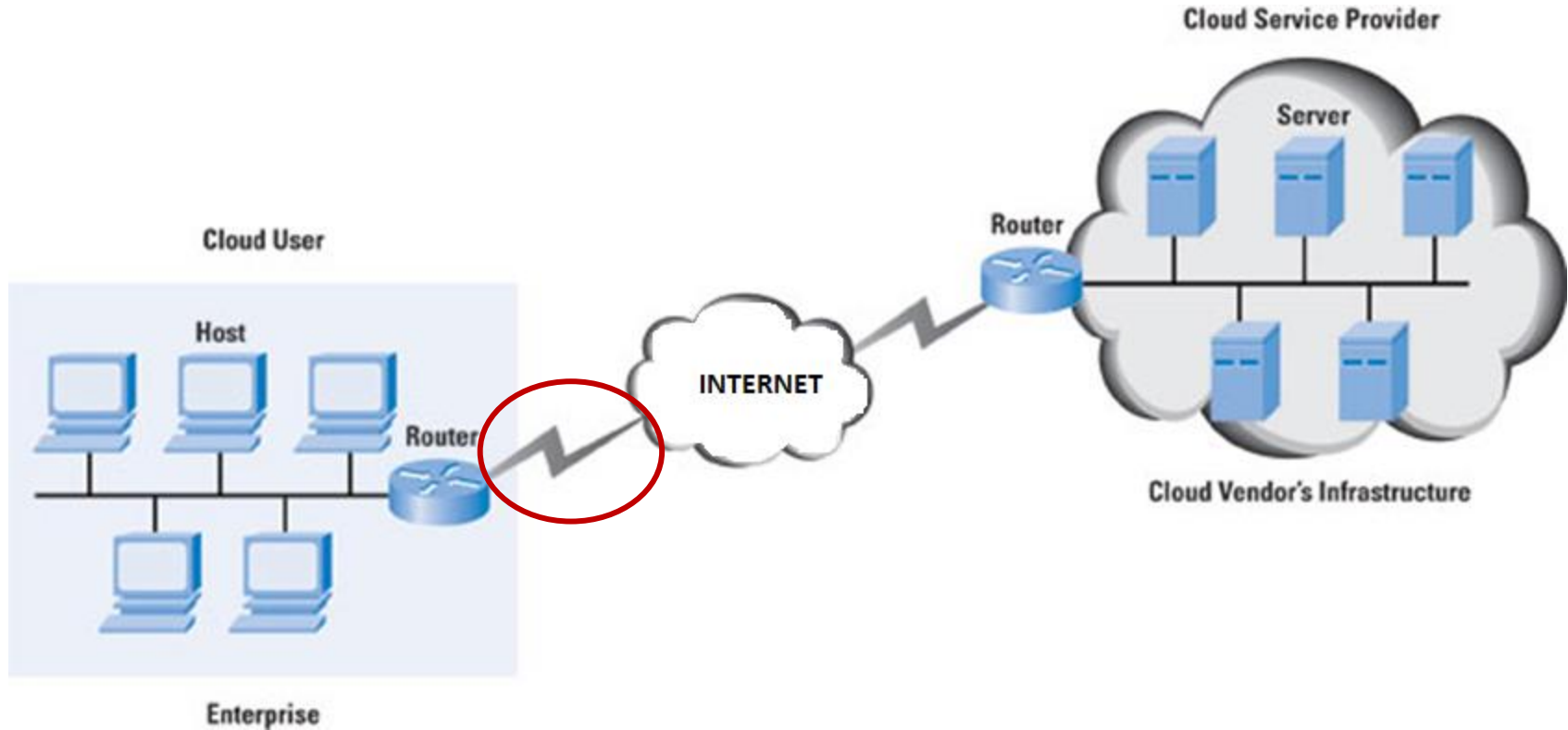
If your target clients are in the same area of your ISP,
there is no problem...



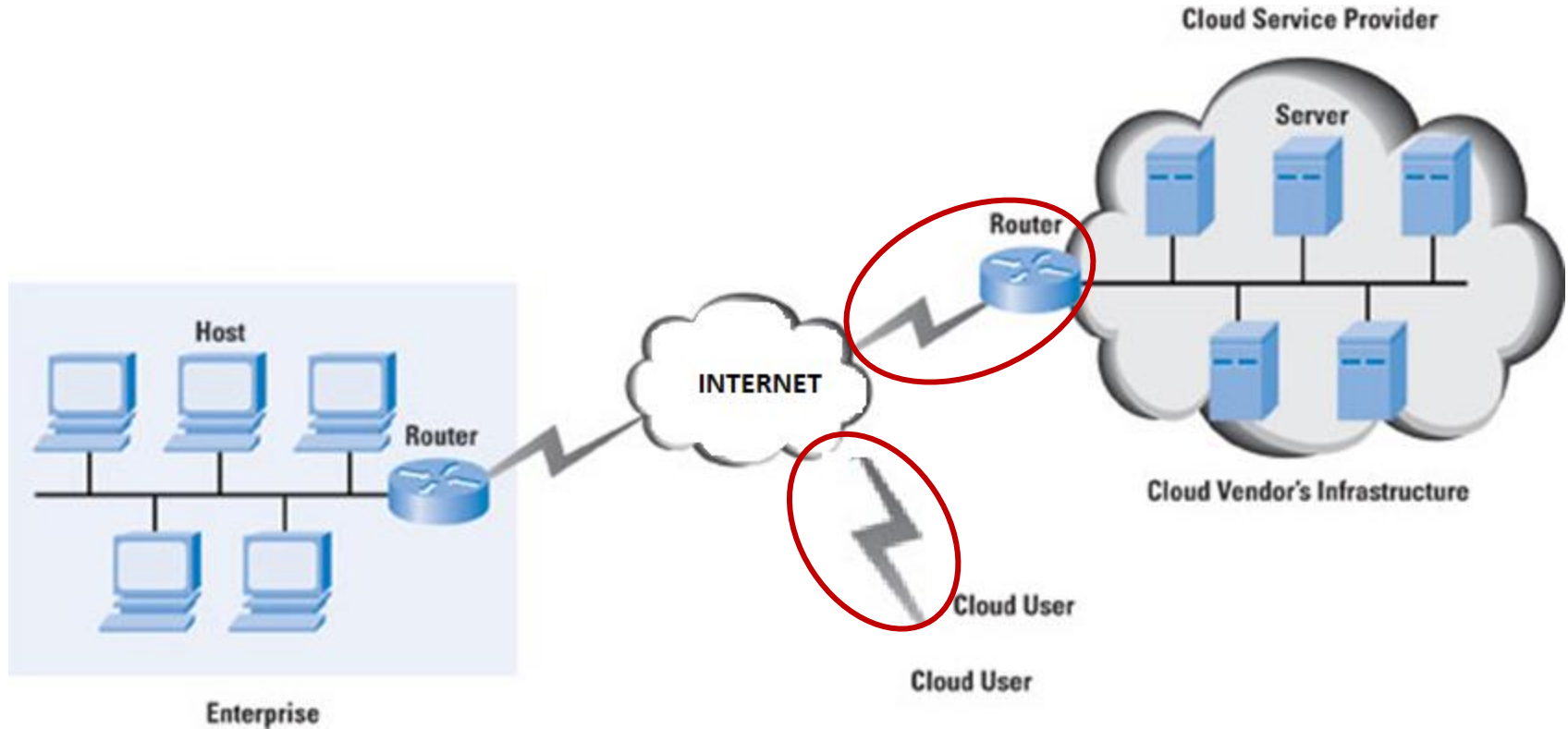
If your target customers are in different AS, then the problem may be more complicated because of Internet



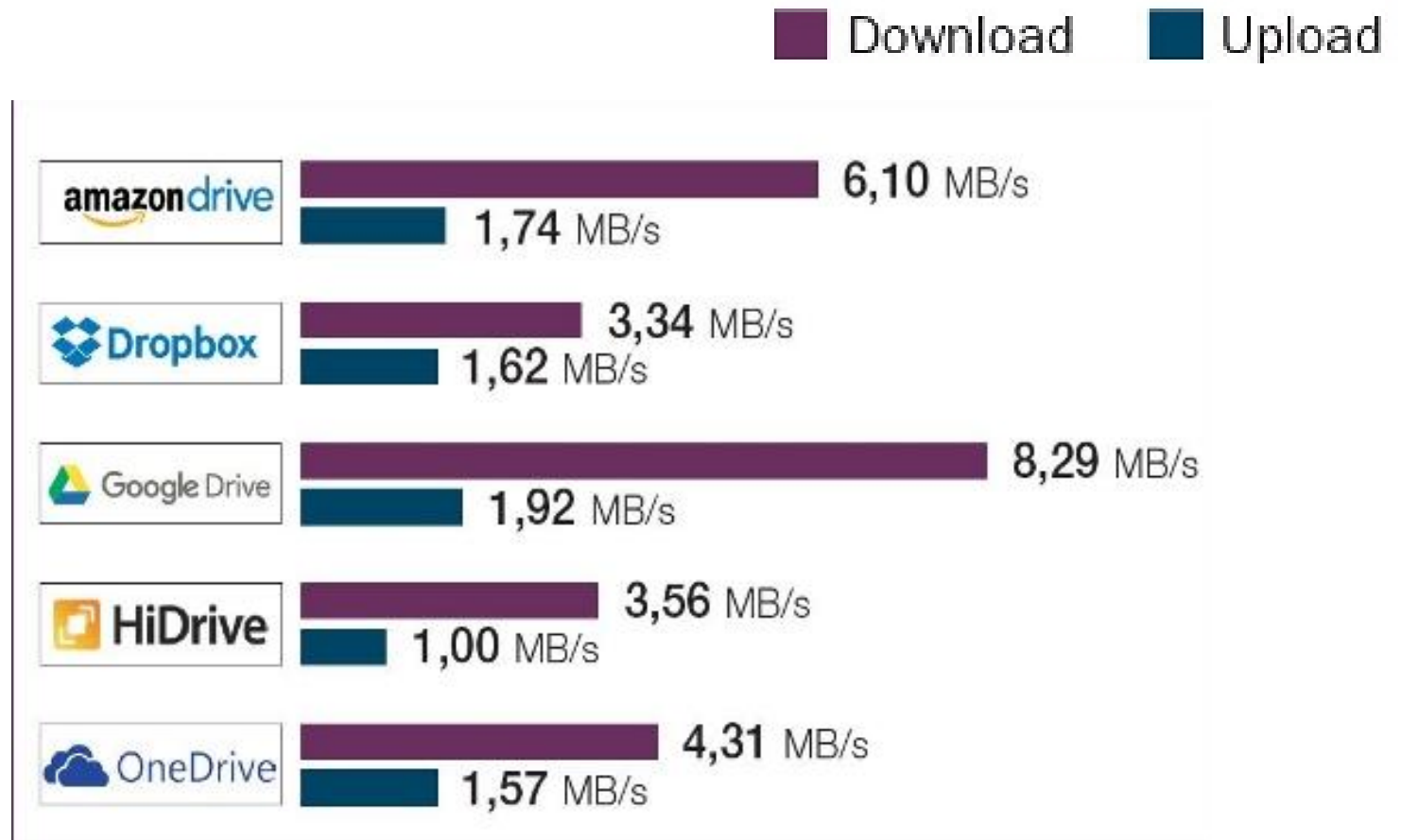
Scenario 1: External datacenter for enterprise use (e.g., email, Office 365, storage, CRM)



Scenario 2: Providing services to external customers



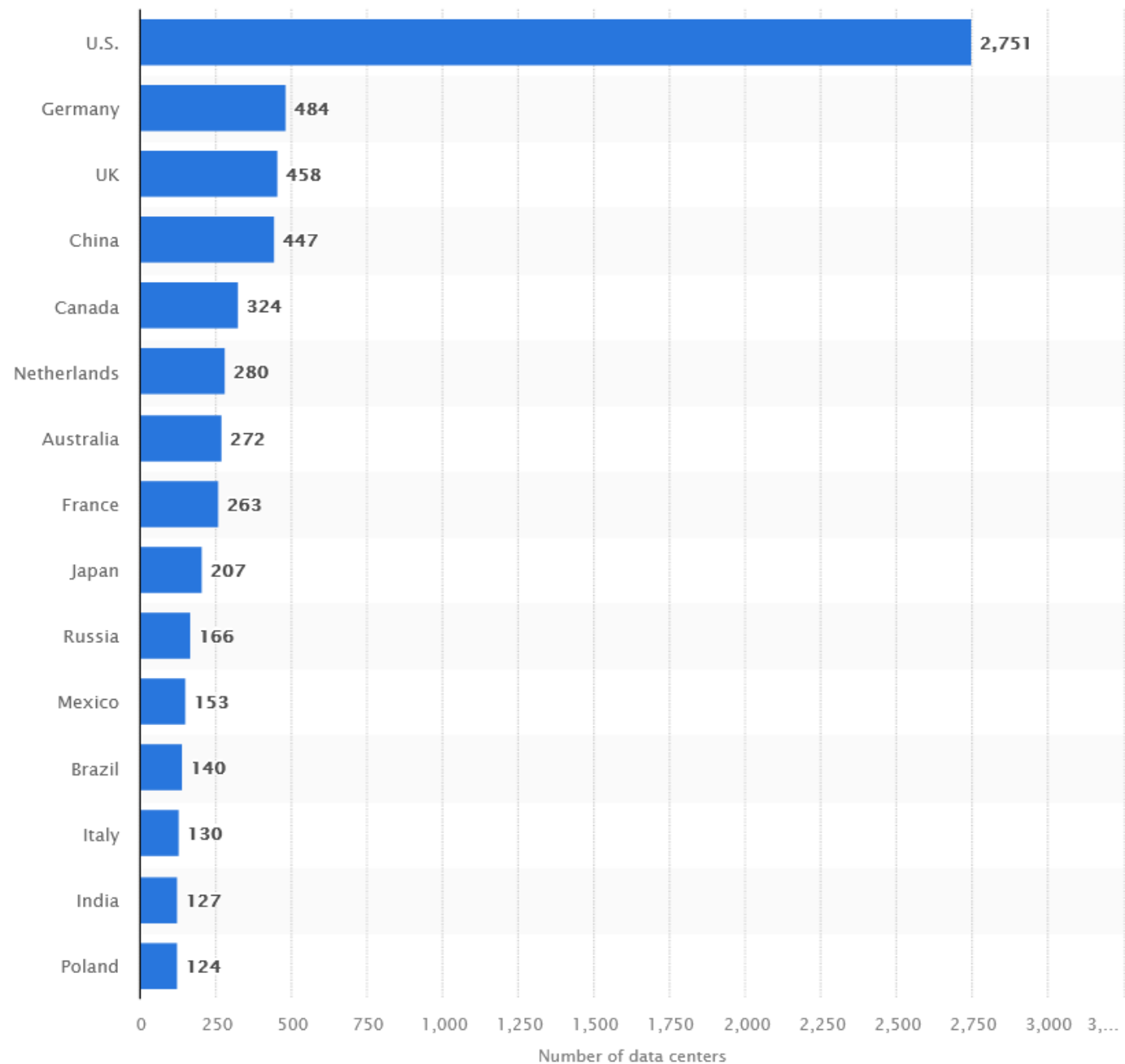
Virtual storage in cloud



Examples of geographical distribution



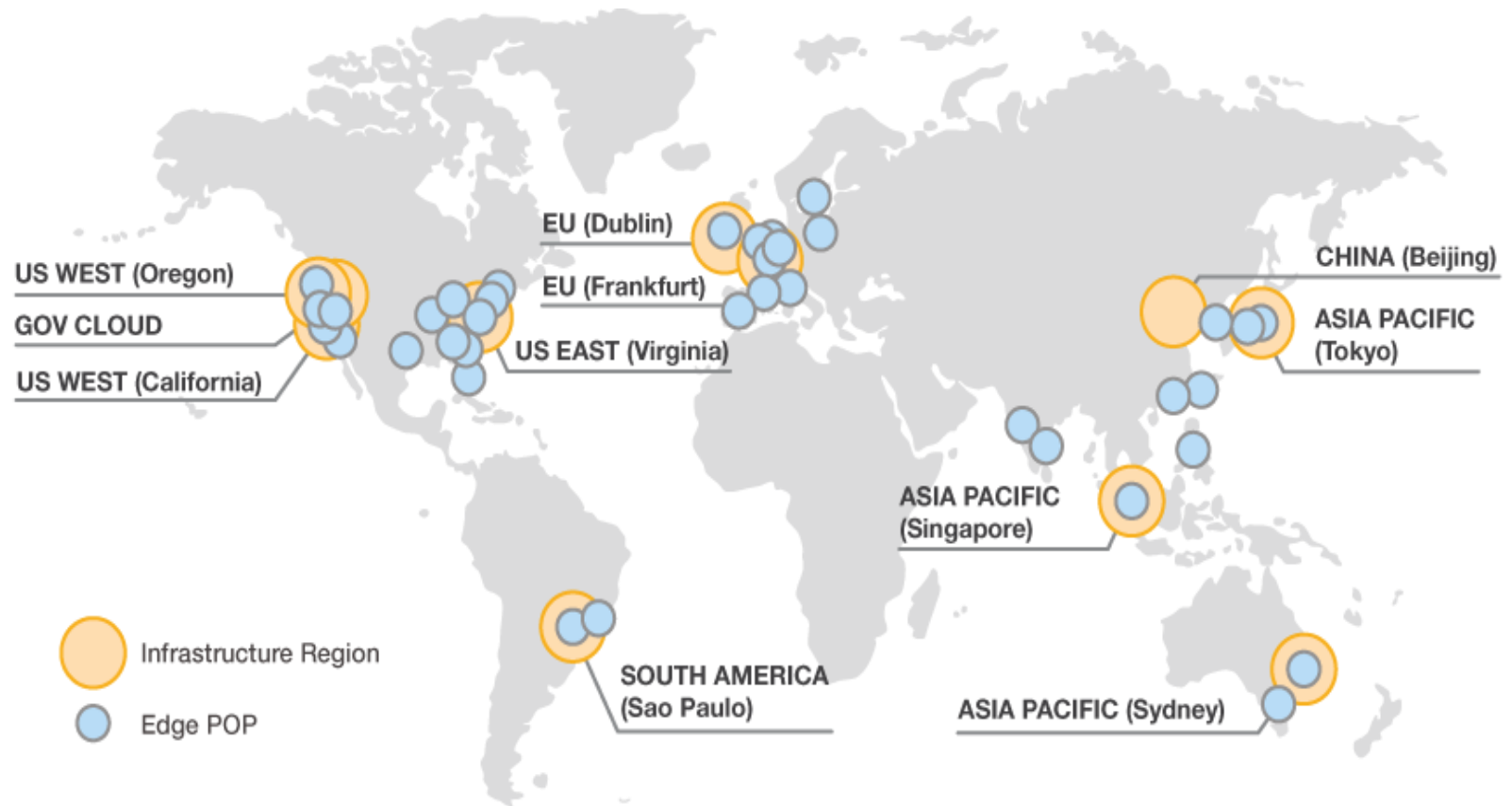
Data centers



Example: Datacenters of Saleforces

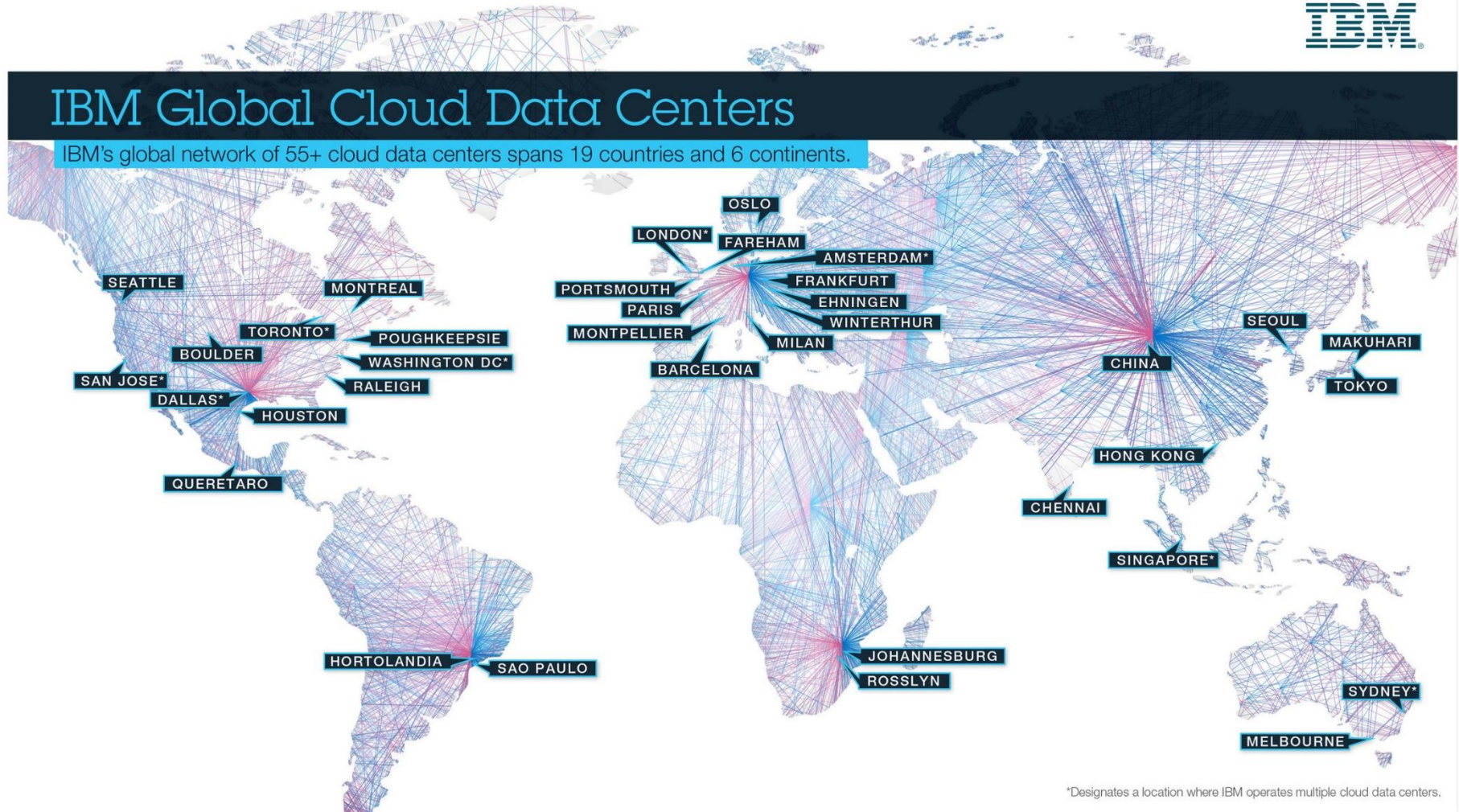


Amazon



IBM Global Cloud Data Centers

IBM's global network of 55+ cloud data centers spans 19 countries and 6 continents.

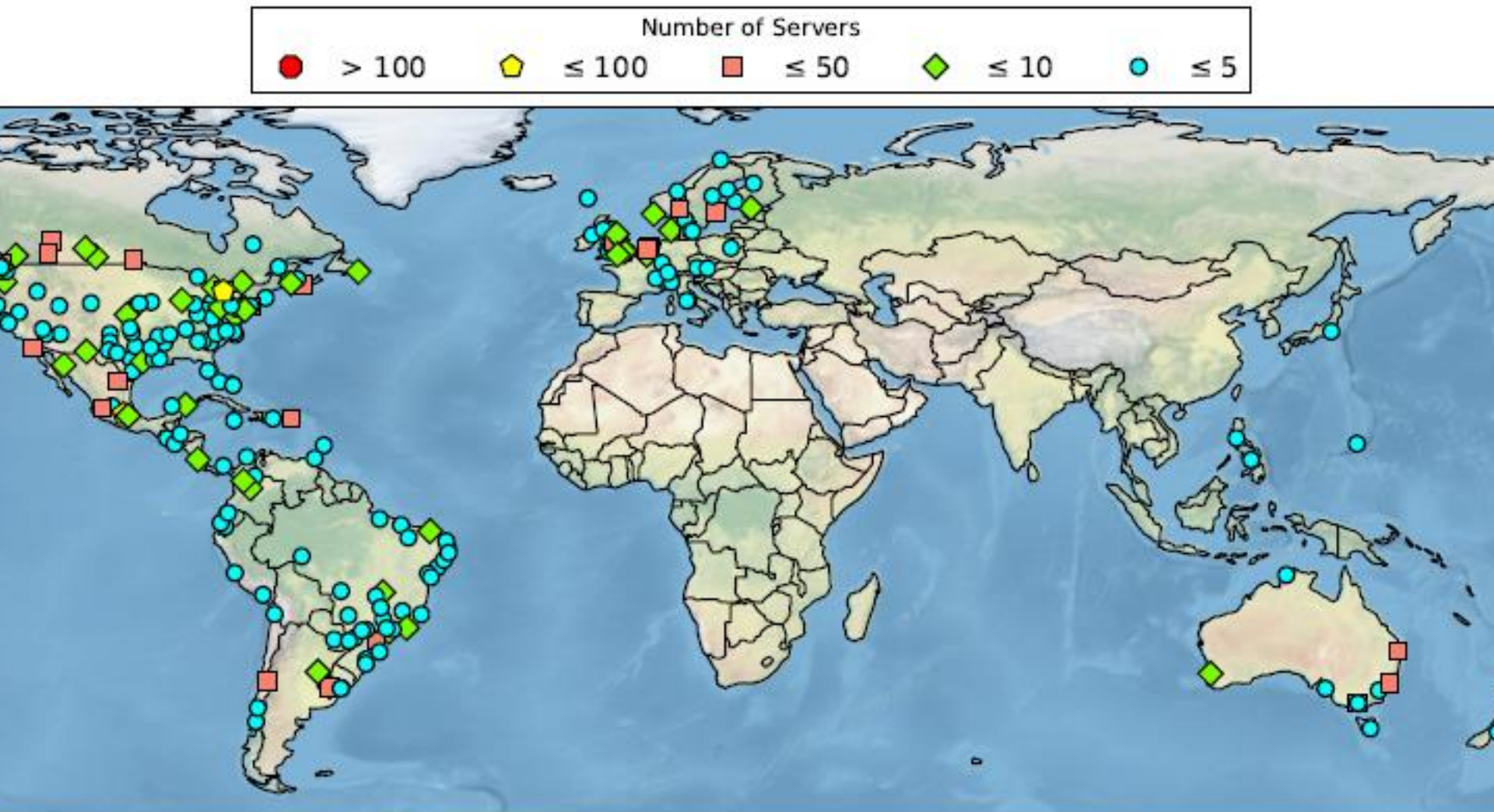


*Designates a location where IBM operates multiple cloud data centers.

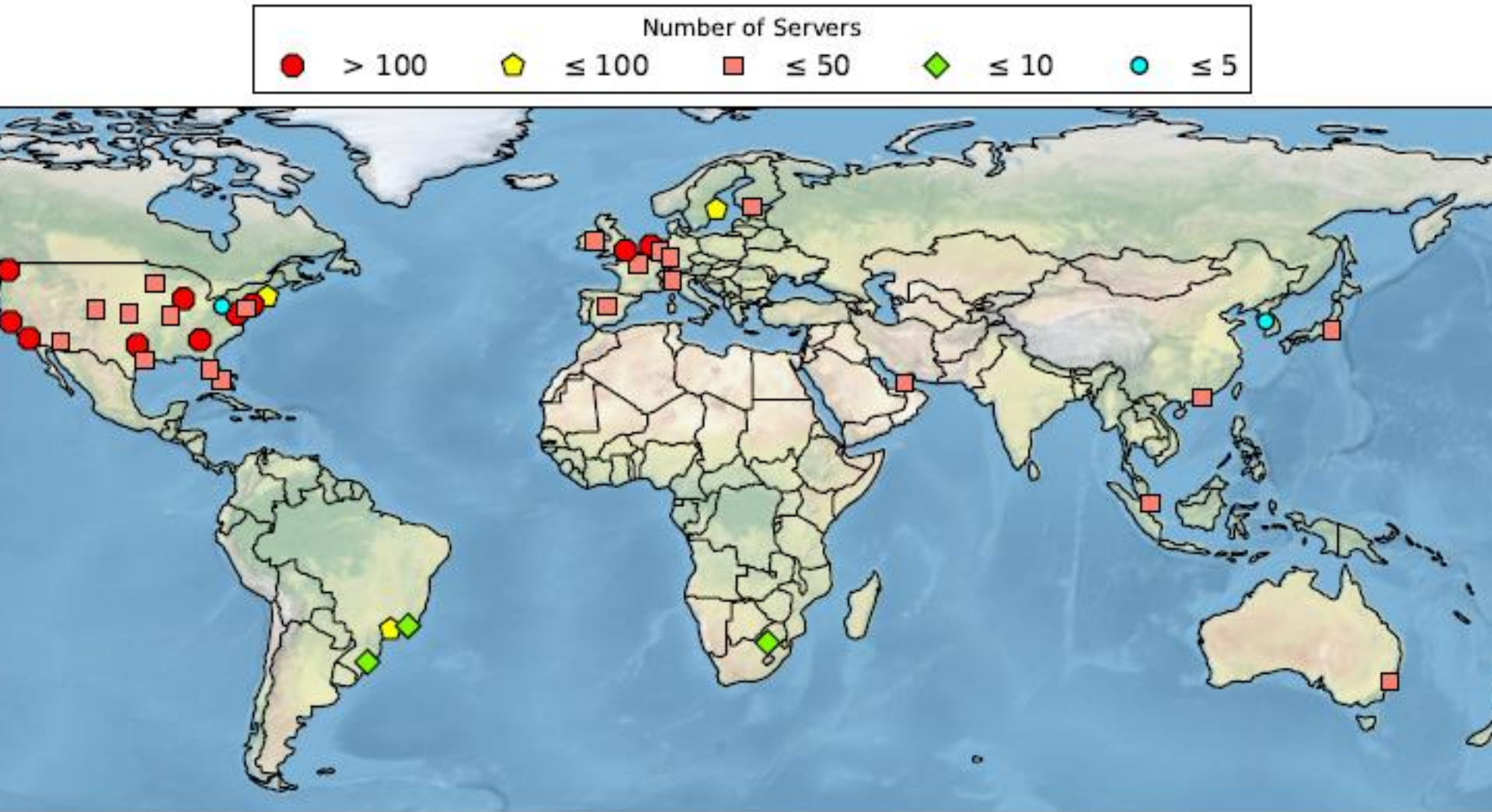
Microsoft Azure



Netflix servers deployed at ISP



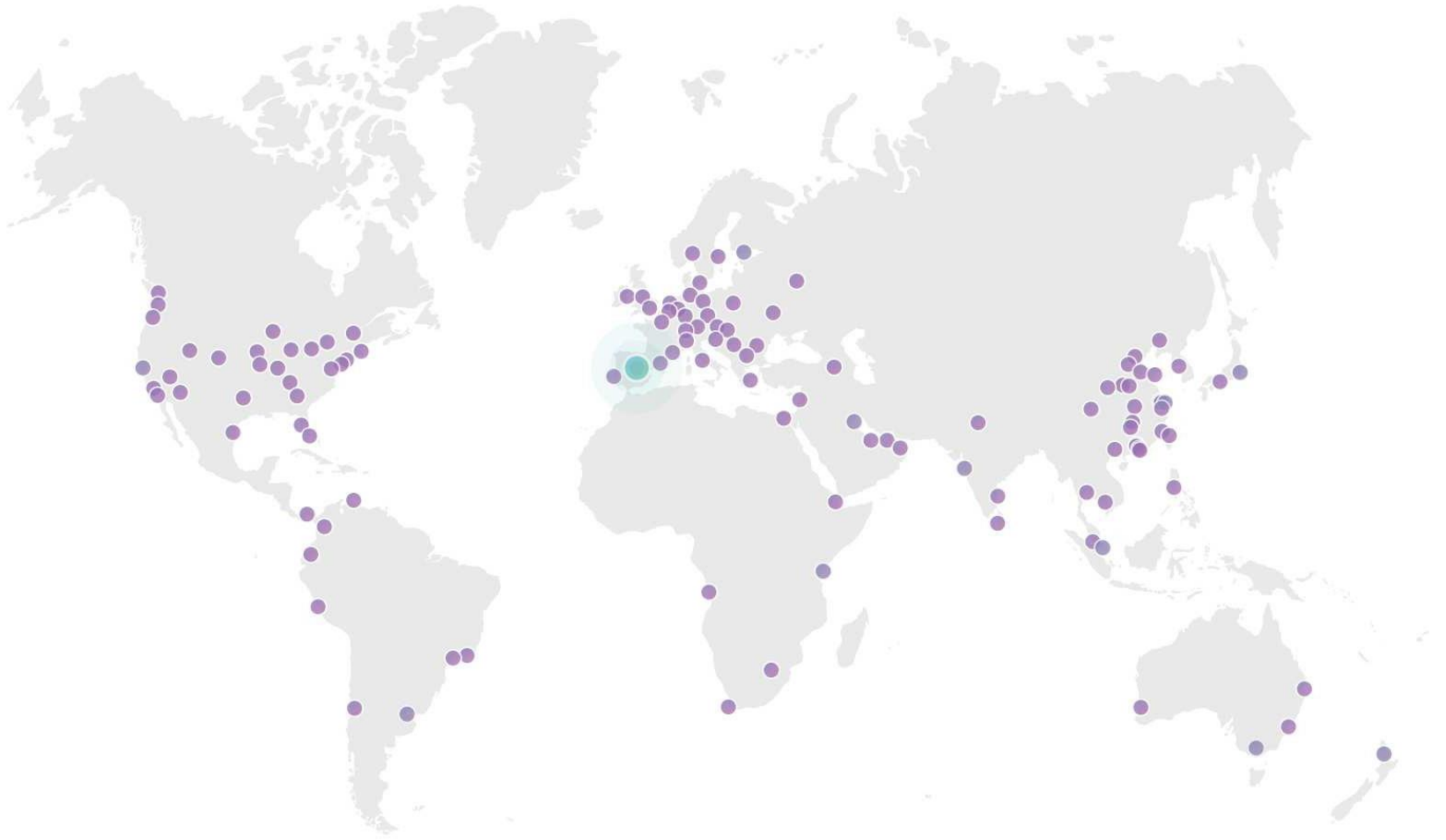
Netflix servers deployed at IXP



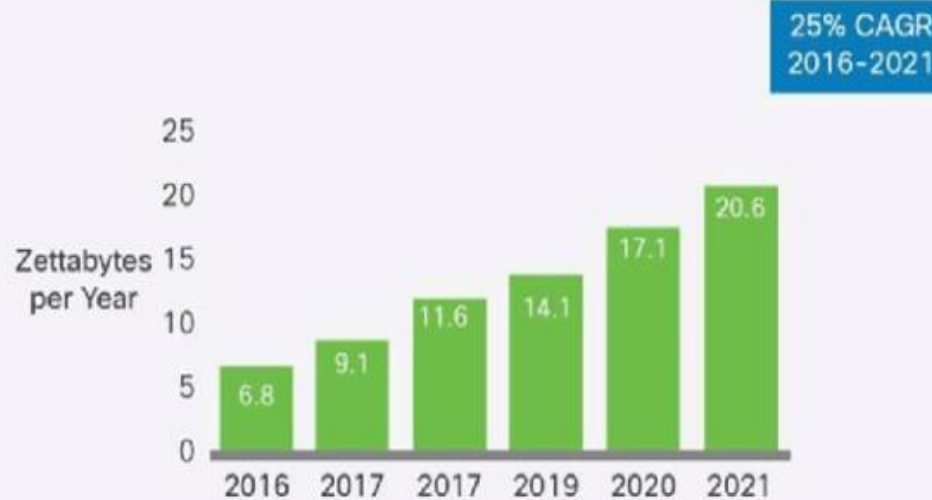
Dropbox edge servers



Cloudflare datacenters



Data Center traffic (*Annual Growth Rate*)



Source: Cisco Global Cloud Index, 2016-2021.

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Effects

- As more software goes to the cloud, the client operating system loses relevance because the browser becomes THE interface
- With the advent of cloud-based productivity software, the importance of MS Office and other desktop software may decrease
- + There could be the possibility of creating and endorsing open standards that everyone can benefit from
- The market dominance that may come as the result of a positive network effect can also have the negative result of creating a single target for ill-intentioned users. If everyone uses Amazon's cloud storage, then that is the only place where someone needs to break into if they want to steal information. The burden on the provider to maintain security and reliability can increase to the point where adding new users actually decreases the total value



The network impact for the *consumer*

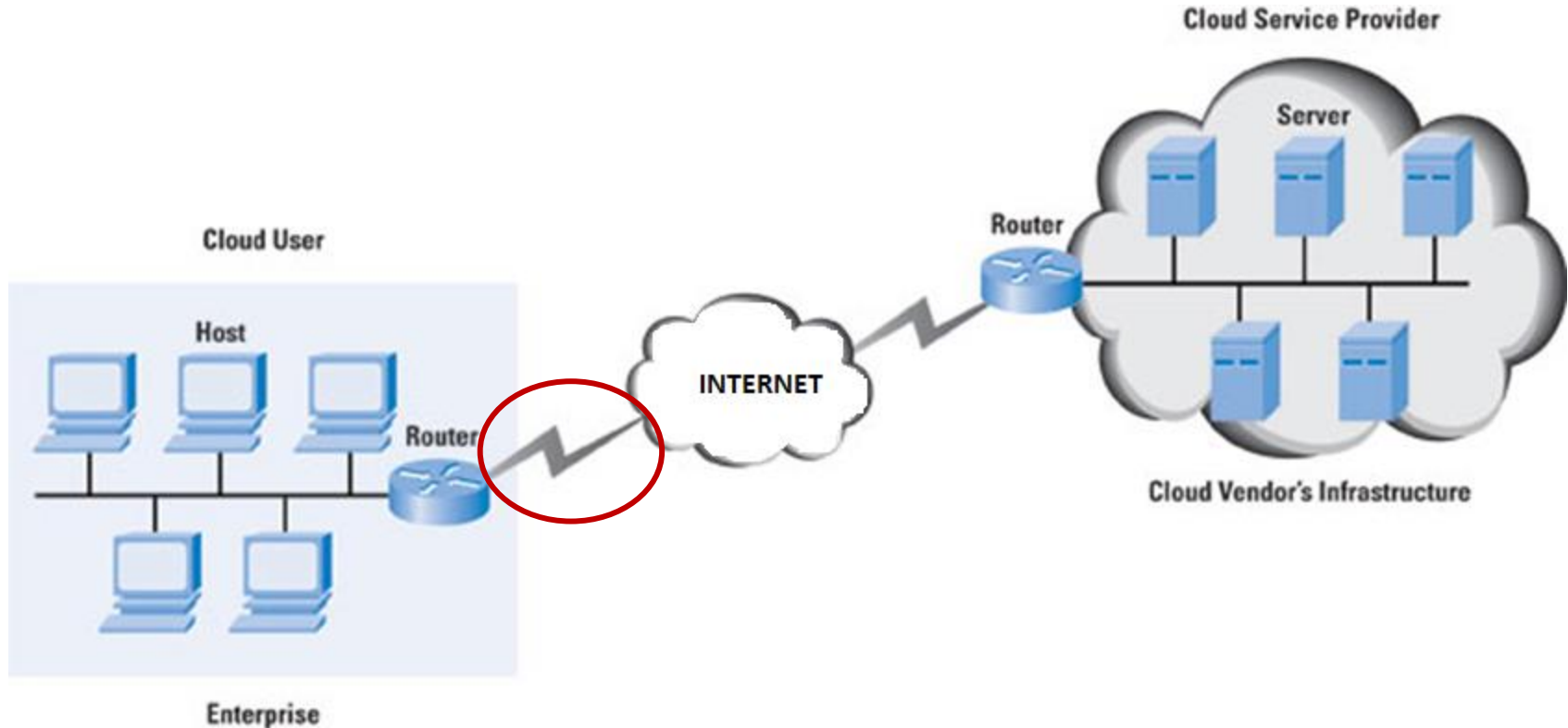
1. Last mile problem

2. Internet traffic

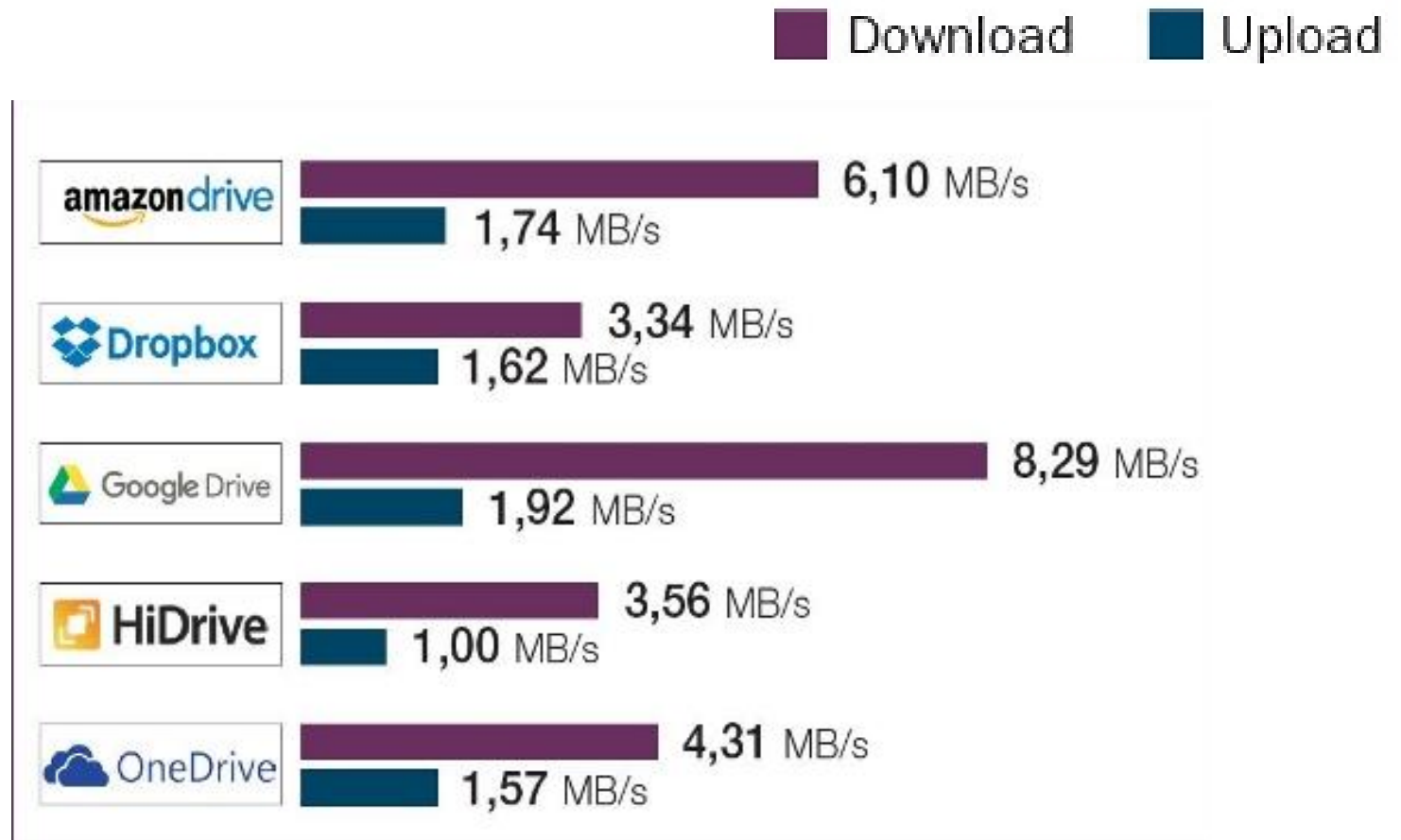
- If you intend to use/integrate external services for your service applications, **bandwidth throughput** to your business locations should be a very significant factor in your decision-making process
 - ➔ If you can choose, then pick the data center that is located in the best connected AS
 - ➔ **If you are not well connected, avoid to put all of your IT services in the cloud** (a good, possible multiple Internet connection is a key parameter to choose a business location for your servers)



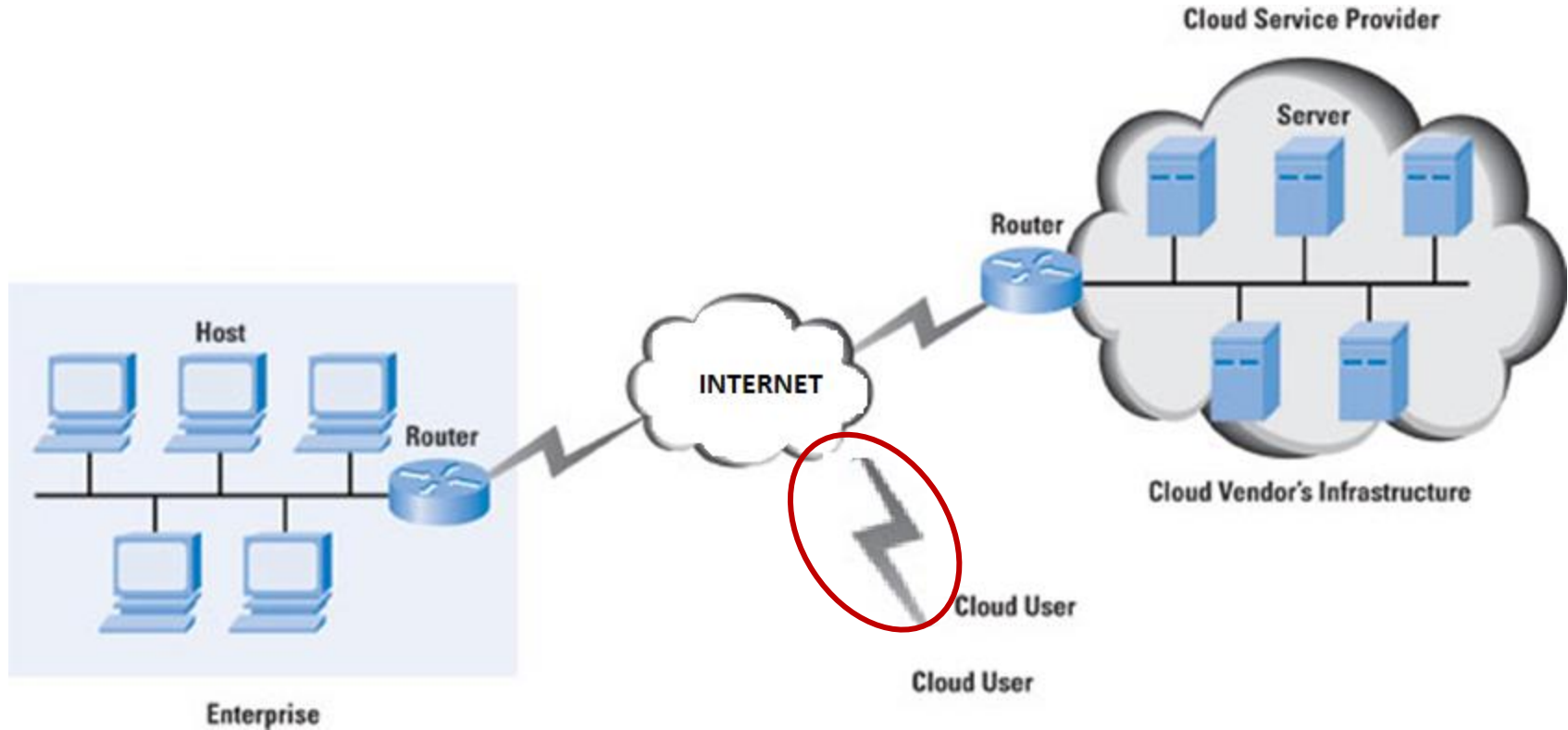
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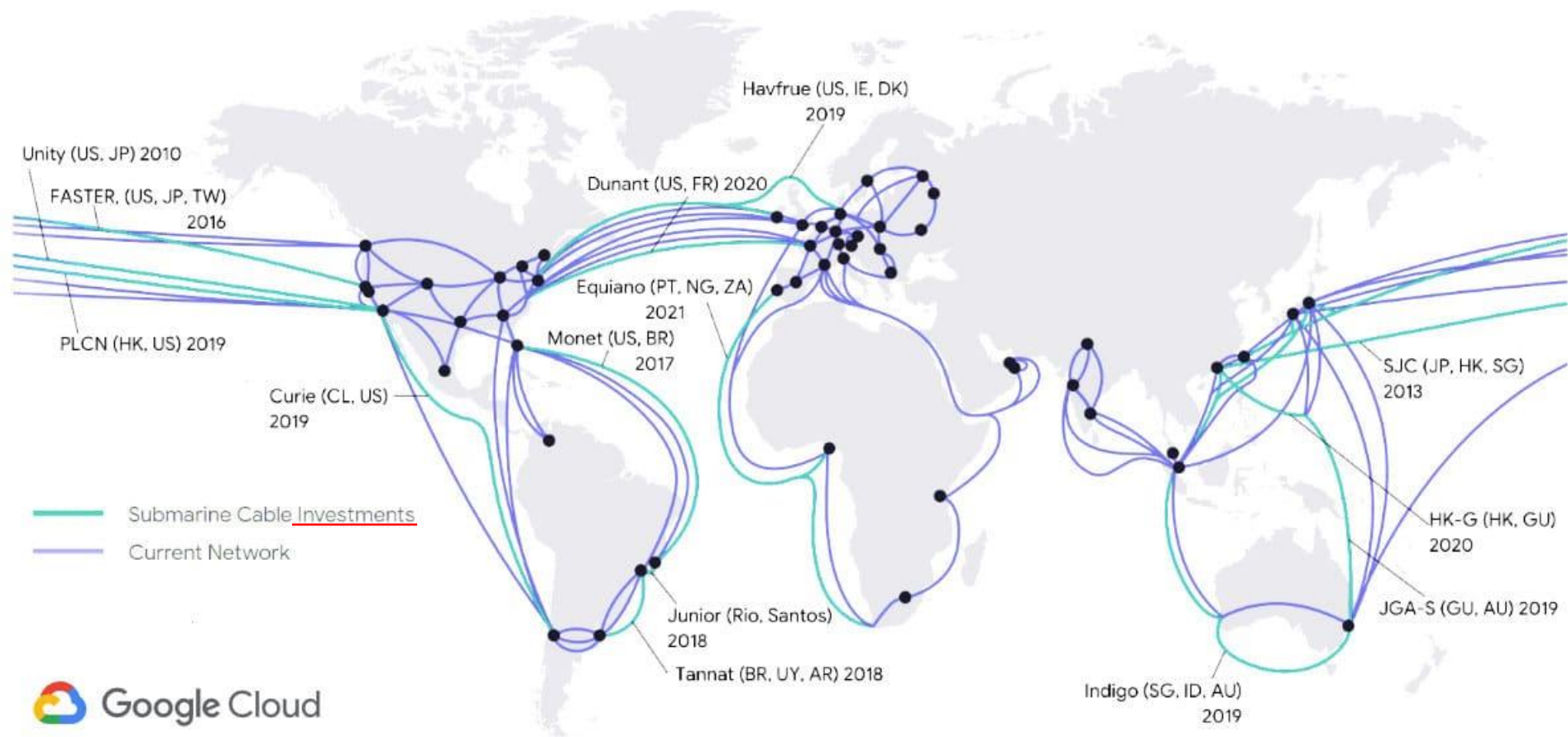
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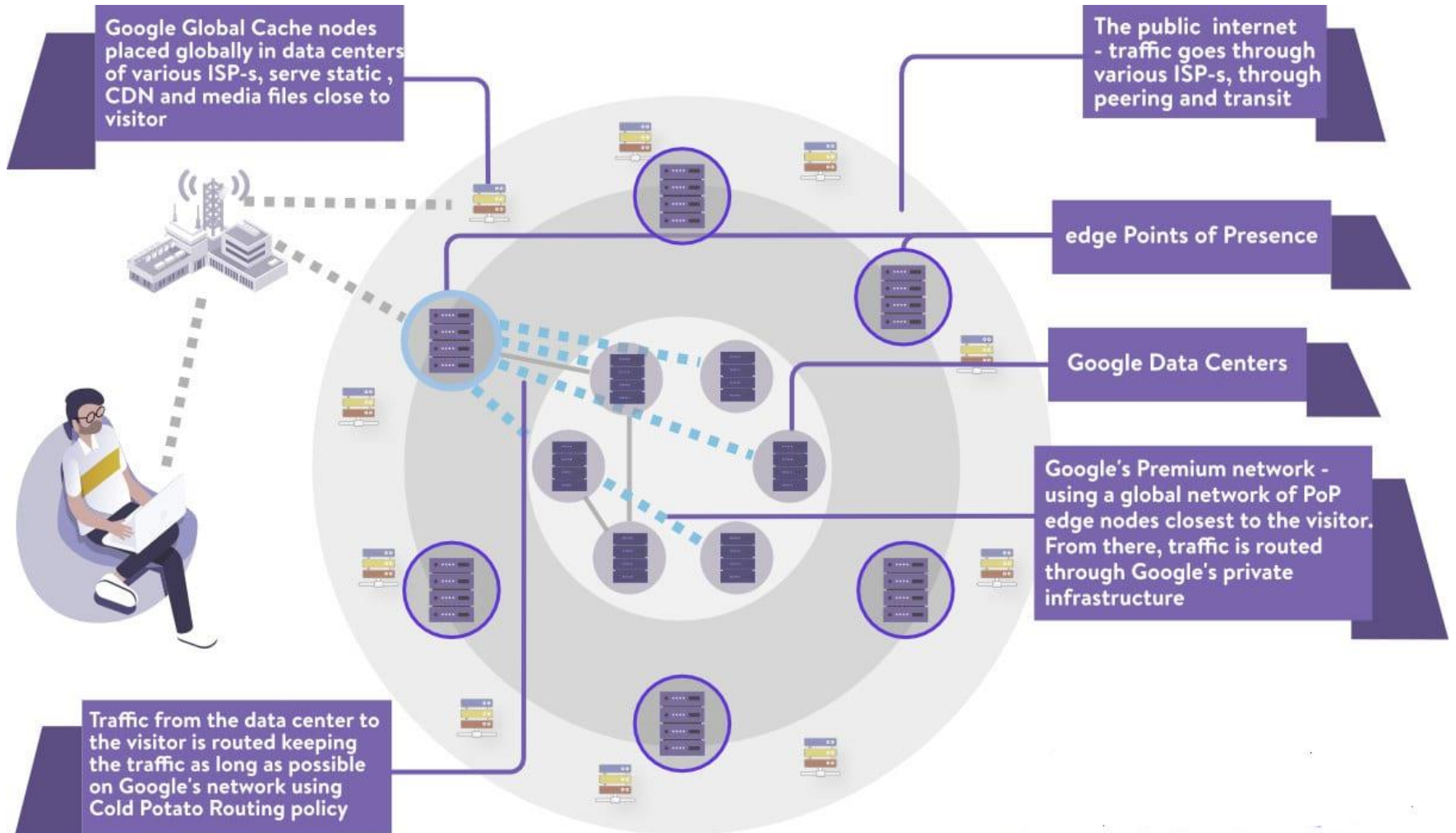
Scenario 2: Cloud for external services



Private WAN networks (*not for everybody*)

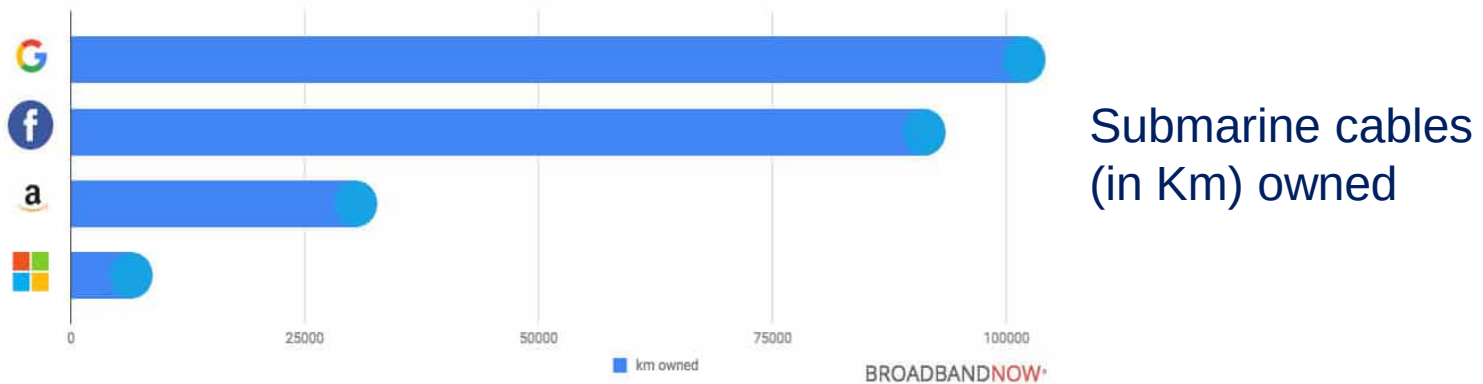


Traffic management in Google



Other cloud providers

- Are following the Google path



- Tend to establish partnerships with Tier-1 and Tier-2 ISPs
- Some cloud providers offer private (direct) connections with their Points of Presence (POP)